

Elevator prob.

Applications of Newton's Laws

- ❑ Newton's first law
- ❑ Newton's second law
- ❑ Newton's third law
- ❑ Frictional forces
- ❑ Applications of Newton's laws
- ❑ Circular Motion



Isaac Newton's work represents one of the greatest contributions to science ever made by an individual.





Force is a vector
Unit of force in S.I.:

$$1N = 1 \frac{kg \cdot m}{s^2}$$

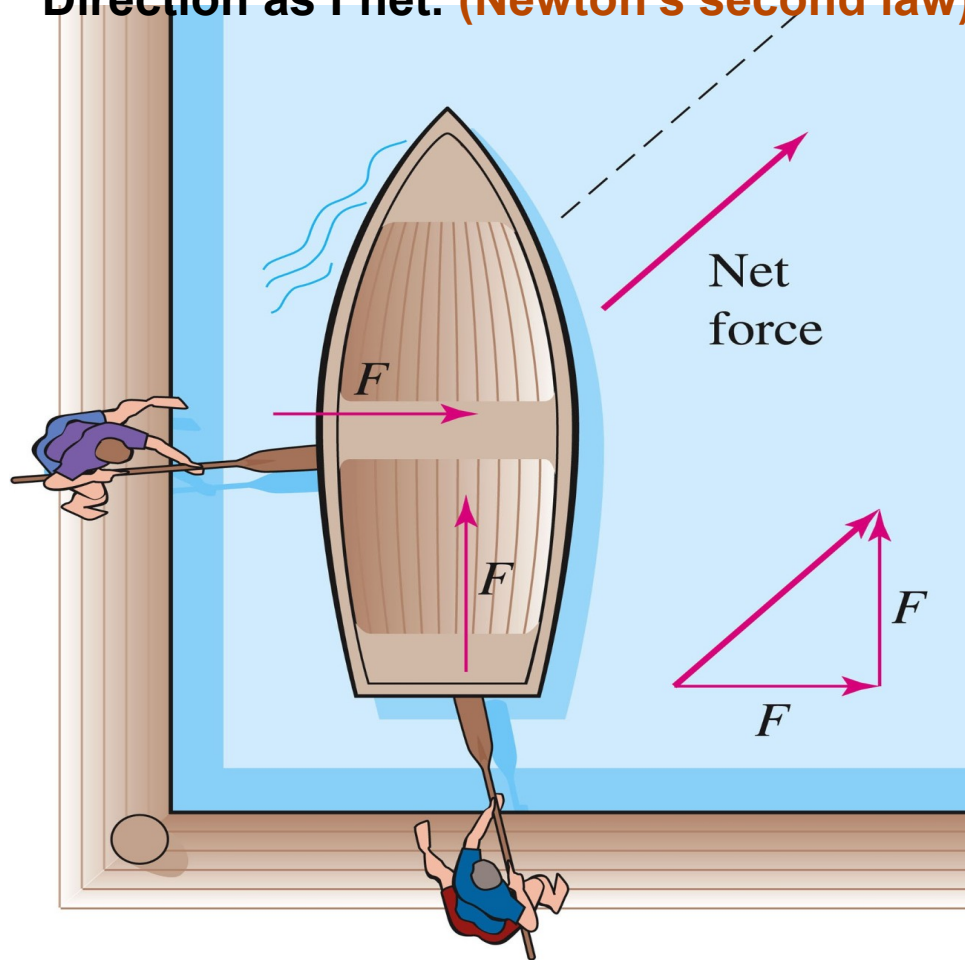
Newton's Laws

- I. If no net **force** acts on a body, then the body's velocity cannot change.
- II. The net **force** on a body is equal to the product of the body's mass and acceleration.
- III. When two bodies interact, the **force** on the bodies from each other are always equal in magnitude and opposite in direction.



Special case: ITS EASY TO ADD VECTORS IF THEY ARE PERPENDICULAR

**In which direction do you think the boat will move ? =
Draw the resultant F_{net} (a vector). The boat will move in the same
Direction as F_{net} . (Newton's second law)**

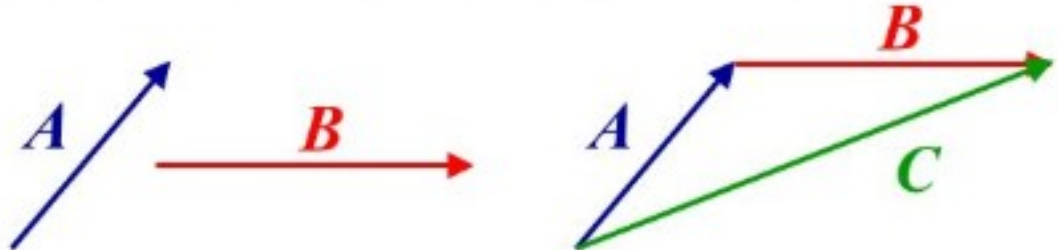


If $|F| = 100\text{N}$
Find R or F_{net}

First draw a free-body
Diagram :attach the tails
Of the forces at the origin.
The boat is reduced to
The origin.

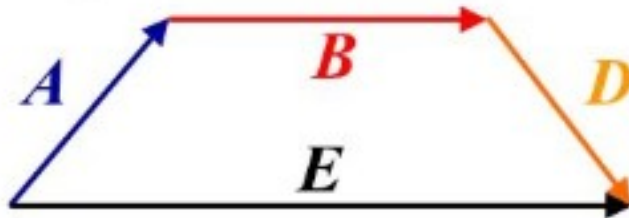
Vector Addition and Subtraction

- For $A + B = C$:



– Vector addition is commutative (order doesn't matter) – try it!

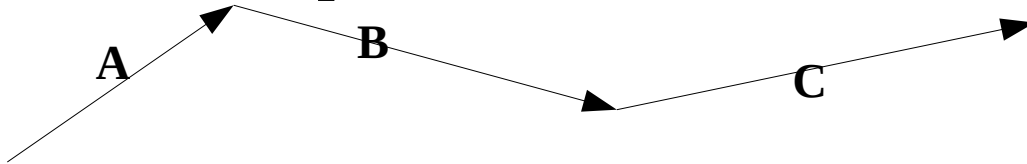
- For multiple vectors $A + B + D = E$:



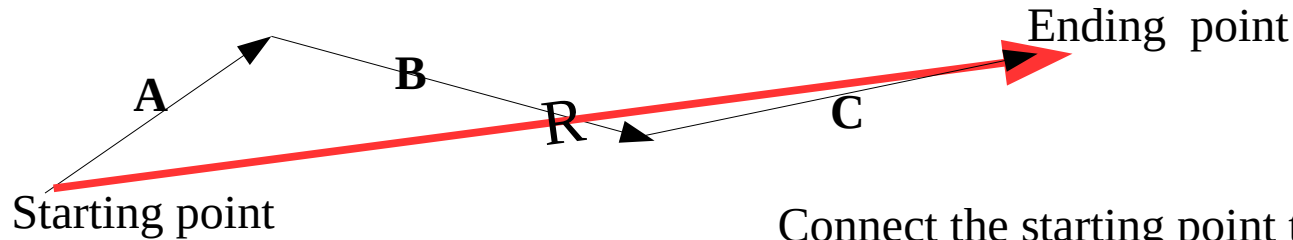
Vector Addition and
Subtraction
Interactive

How to add vectors: method head to tail (tail to head)

Example: add the vectors A, B, C

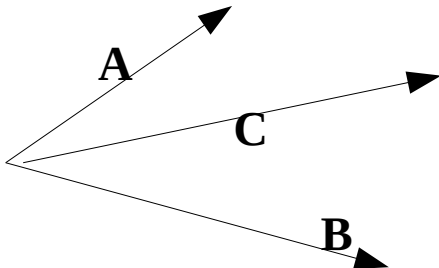


$A+B+C = R$ R is the vector sum (or resultant)

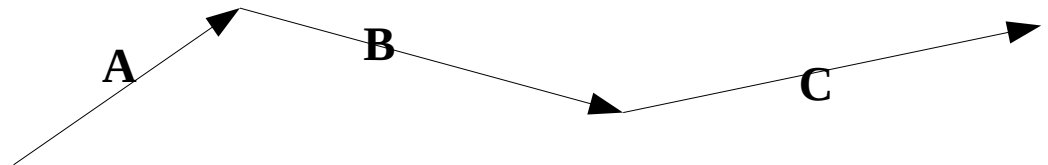


Connect the starting point to ending point
The vectors are tail to head (head to tail)

If the vectors are not tail to head then move them tail to head

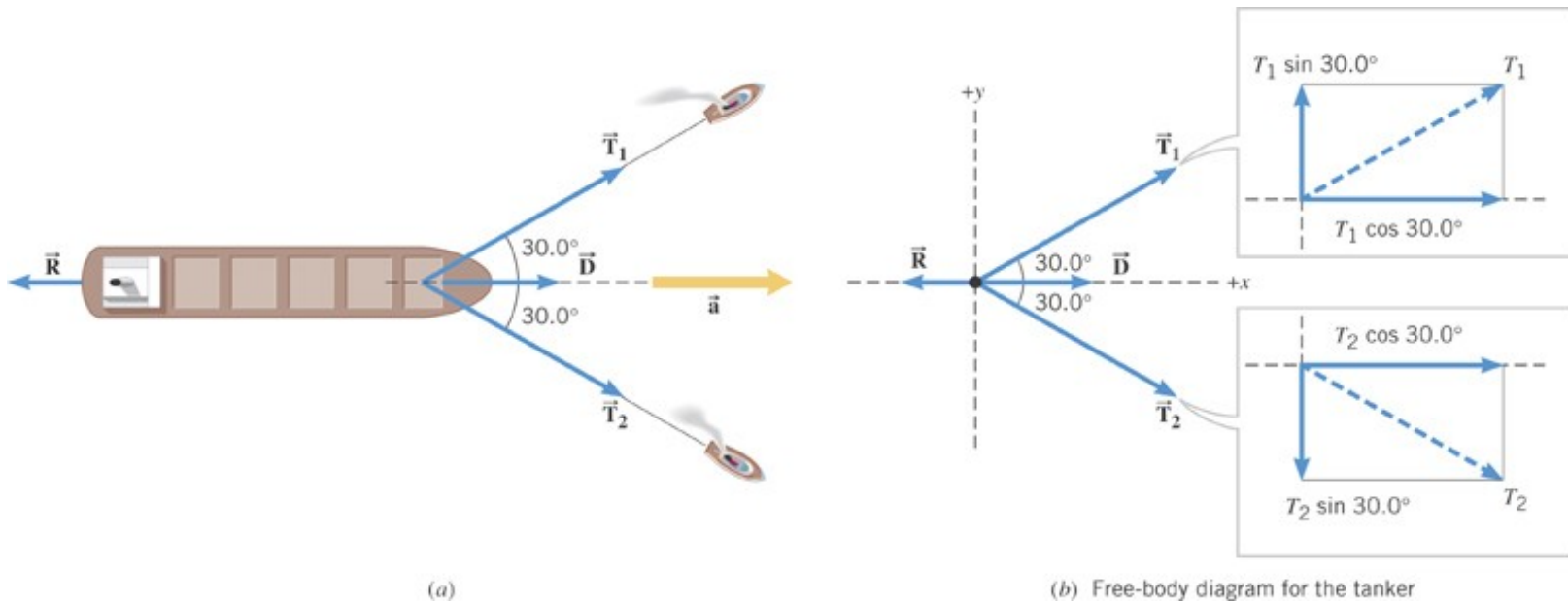


Move the tail of B on head of A and the tail of C on head of B



Try: <http://www.physicsclassroom.com/class/vectors/Lesson-1/Vector-Addition>

**Suppose we know R , D , mass, acceleration a
What is T_1 ? T_2 ?**

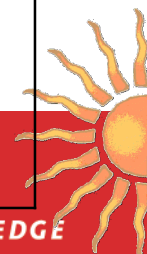


The acceleration is along the x axis so $a_y = 0$



4.12 Nonequilibrium Application of Newton's Laws of Motion

Force	x component	y component
$\vec{T}_1$	$+T_1 \cos 30.0^\circ$	$+T_1 \sin 30.0^\circ$
$\vec{T}_2$	$+T_2 \cos 30.0^\circ$	$-T_2 \sin 30.0^\circ$
$\vec{D}$	$+D$	0
$\vec{R}$	$-R$	0



4.12 Nonequilibrium Application of Newton's Laws of Motion

$$\sum F_y = +T_1 \sin 30.0^\circ - T_2 \sin 30.0 = 0$$

$$\Rightarrow T_1 = T_2$$

$$\begin{aligned}\sum F_x &= +T_1 \cos 30.0^\circ + T_2 \cos 30.0 + D - R \\ &= ma_x\end{aligned}$$



4.12 Nonequilibrium Application of Newton's Laws of Motion

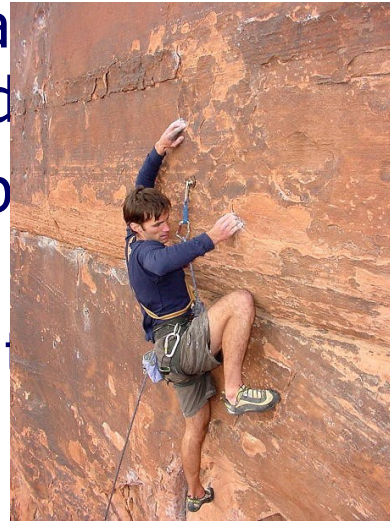
$$T_1 = T_2 = T$$

$$T = \frac{ma_x + R - D}{2 \cos 30.0^\circ}$$

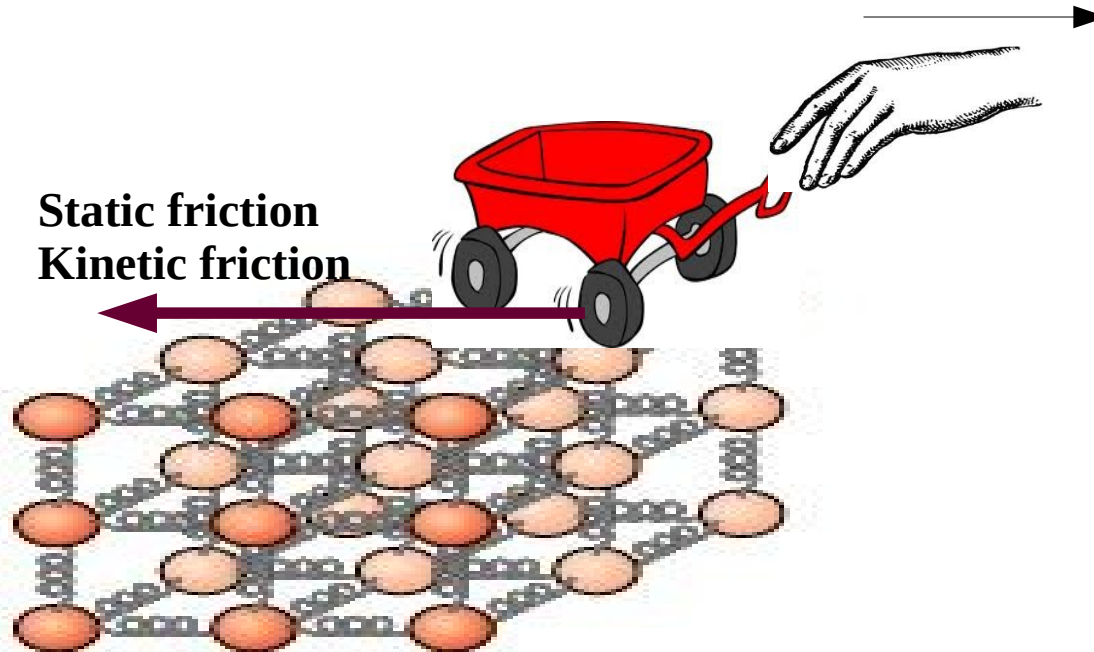


Forces of Friction: f

- ❑ When an object is in motion on a surface or through a viscous medium, there will be a resistance to motion. This resistance is called friction.
- ❑ This is due to the interactions between the object and its environment
- ❑ We will be concerned with two types of friction
 - Force of static friction: f_s
 - Force of kinetic friction: f_k
- ❑ Direction: opposite the direction of the intended motion
 - If moving: in direction opposite the velocity
 - If stationary, in direction of the vector sum of other forces

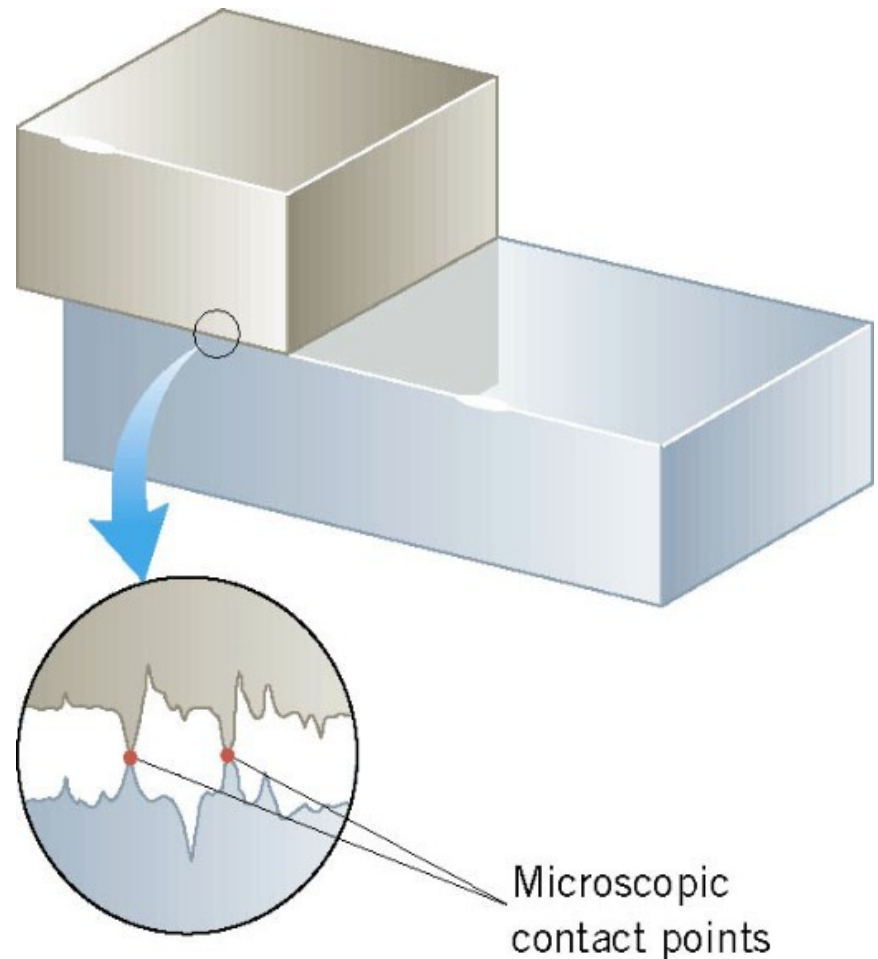


Frictional forces

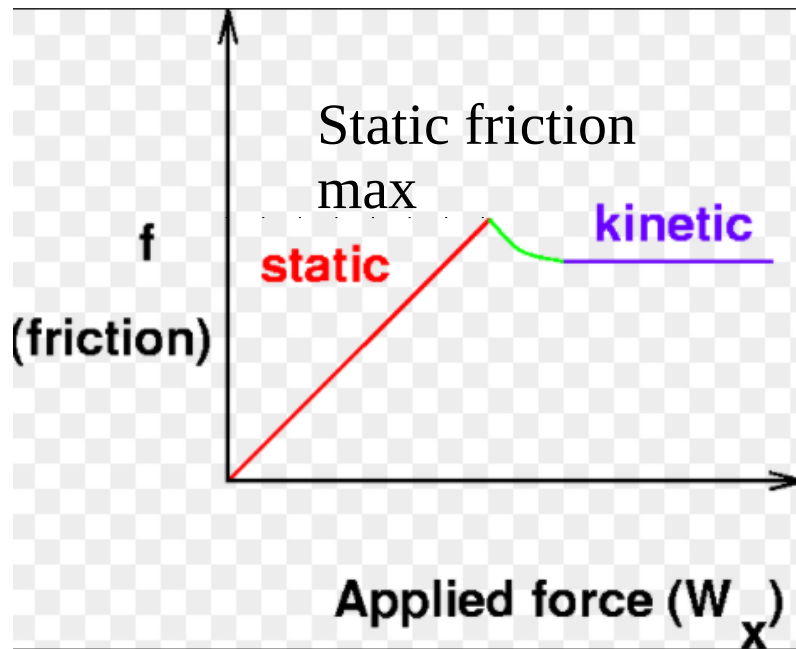


<http://phet.colorado.edu/en/simulation/forces-and-motion-basics>

When an object is in contact with a surface there is a force acting on that object. The component of this force that is parallel to the surface is called the ***frictional force***.

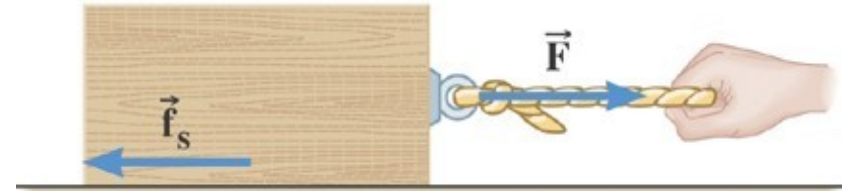


When the two surfaces are not sliding across one another the friction is called ***static friction***.



No movement

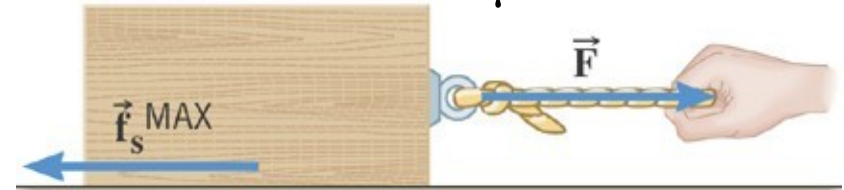
(a)



No movement

(b)

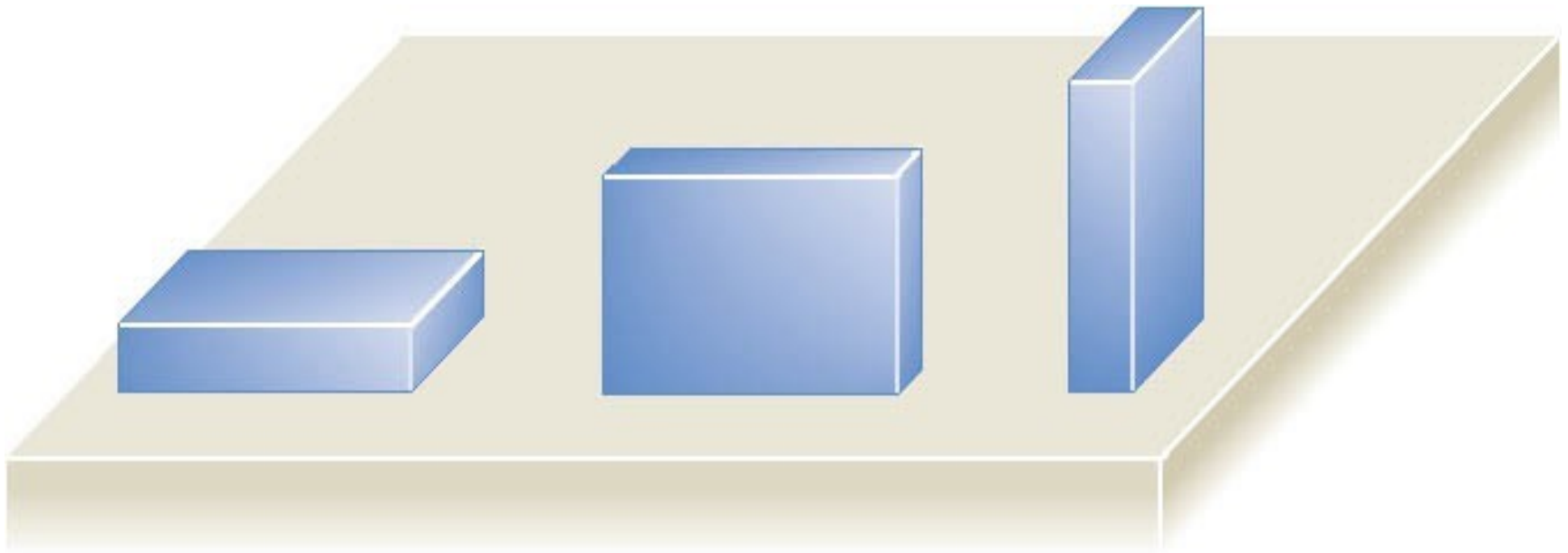
Static friction max = $\mu \times \text{Normal}$



When movement just begins

(c)

Note that the magnitude of the frictional force does not depend on the contact area of the surfaces.



Static friction opposes the *impending* relative motion between two objects.

Kinetic friction opposes the relative sliding motion motions actually does occur.

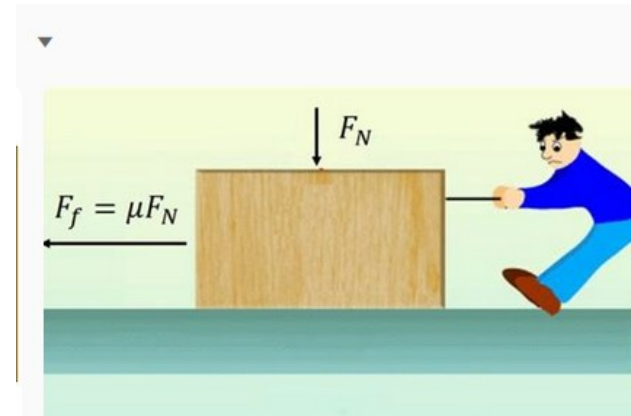
$$f_k = \mu_k F_N$$



$0 < \mu < 1$ is called the coefficient of kinetic friction

Approximate Coefficients of Friction

	Kinetic	Static
Rubber on concrete (dry)	0.68	0.90
Rubber on concrete (wet)	0.58	
Rubber on asphalt (dry)	0.67	0.85
Rubber on asphalt (wet)	0.53	
Rubber on ice	0.15	
Waxed ski on snow	0.05	0.14
Wood on wood	0.30	0.42
Steel on steel	0.57	0.74
Copper on steel	0.36	0.53
Teflon on Teflon	0.04	

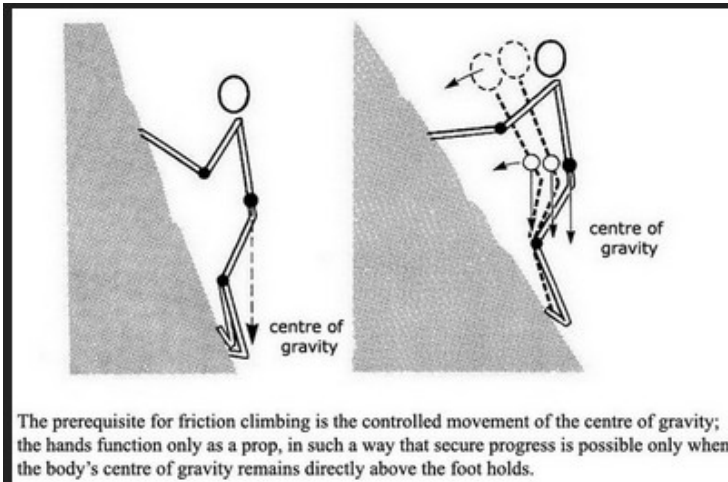


Physics of climbing

→ smearing/edging = use static friction.

Friction = μ weight . To increase friction, increase μ (special rubber shoes)
And increase the weight by standing on one foot . Noise above the foot
So the center of mass is above shoe.

Power increases the coefficient μ .



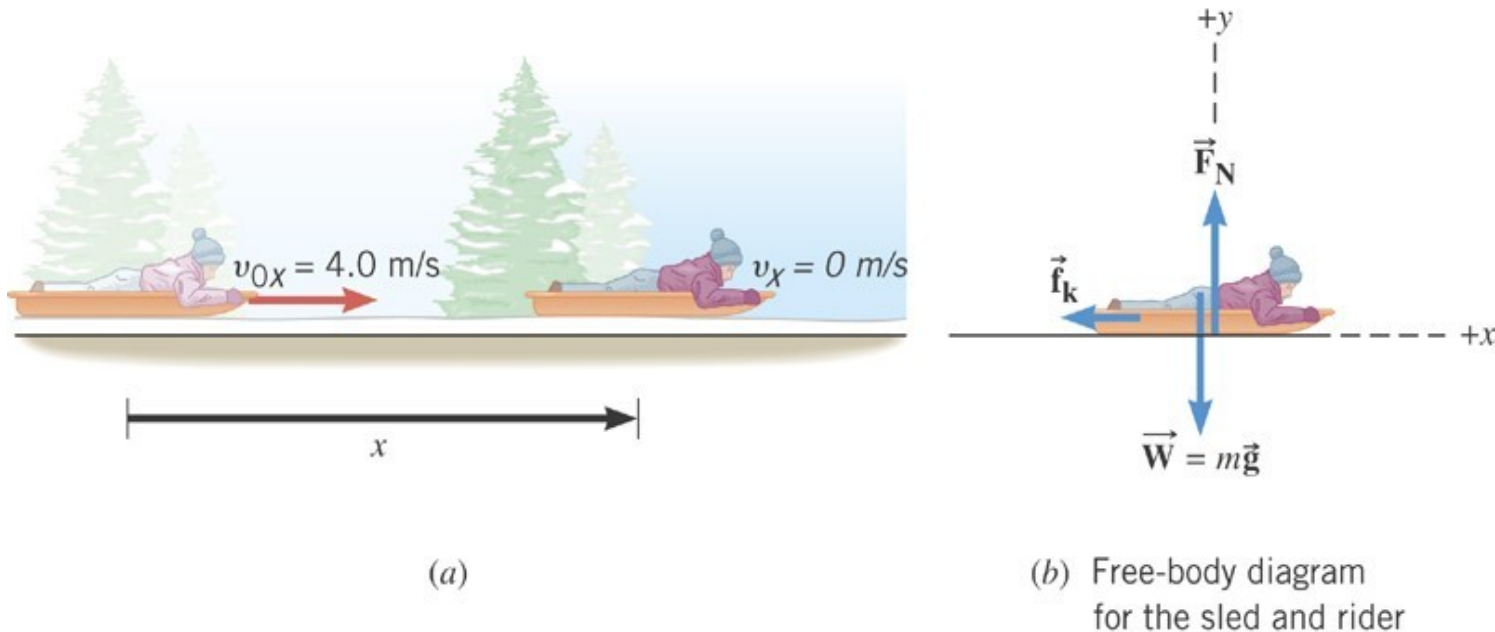
Push on legs first before pulling with Arm (more muscles in legs)

<https://www.youtube.com/watch?v=h9k96NWFn08>



<https://www.youtube.com/watch?v=01vz0g64qQo&t=6s>

Textbook 5.13
5.15



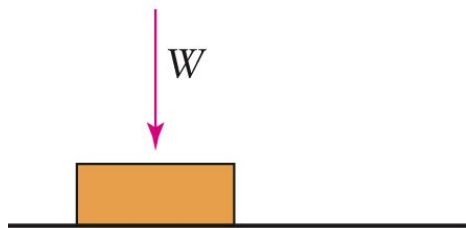
The sled comes to a halt because the kinetic frictional force opposes its motion and causes the sled to slow down.

Suppose the coefficient of kinetic friction is 0.05 and the total mass is 40kg. What is the kinetic frictional force?

□ A block of wood resting on an inclined plane is a simple system that relies on static friction.

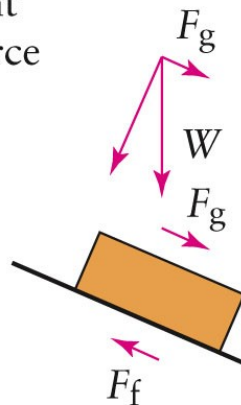
- If the plane is horizontal, there is no friction.
- When the plane is tilted a small amount, friction acts to keep the block from sliding down.

F_g = component of weight
(or gravitational) force
 F_f = force due to friction



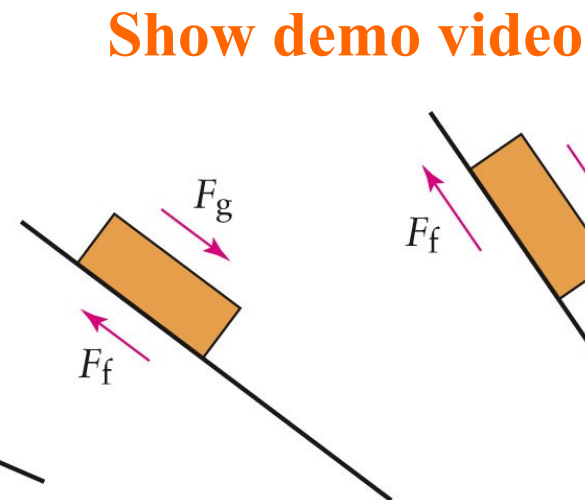
(a)

$$F_g = F_f = 0$$



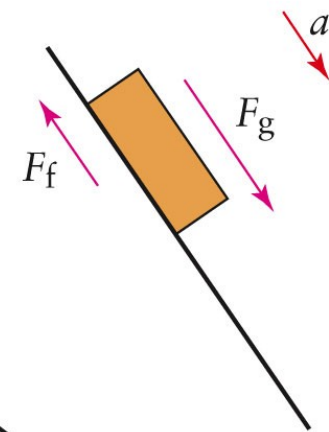
(b)

$$F_g = F_f$$



(c)

$$F_g = F_f$$



(d)

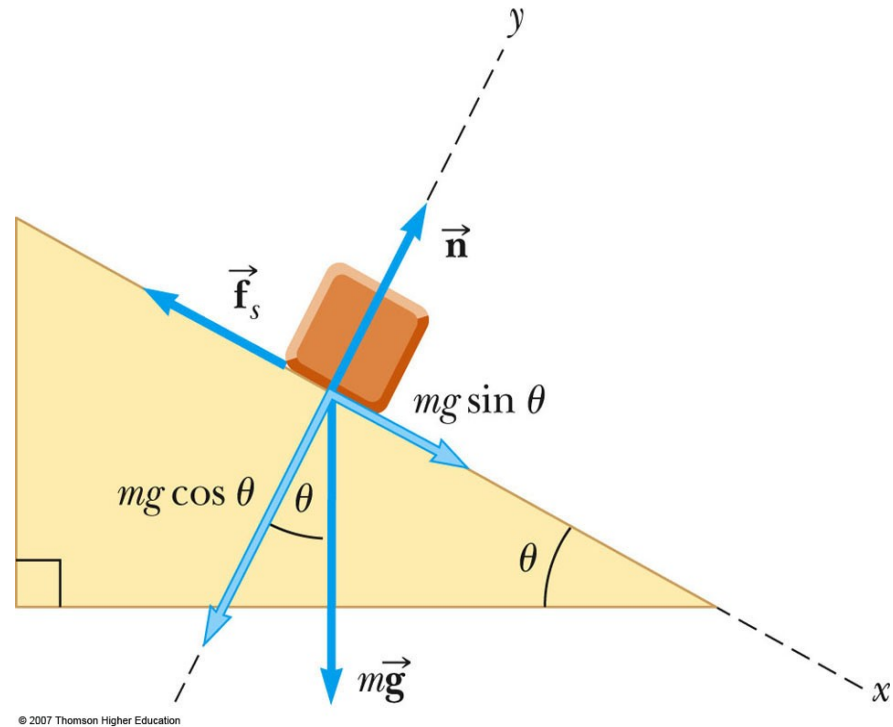
$$F_g > F_{f \text{ max}}$$

Show demo video



Inclined Plane

- Suppose a block with a mass of 2.50 kg is resting on a ramp. If the coefficient of static friction between the block and ramp is 0.350, what maximum angle can the ramp make with the horizontal before the block starts to slip down?



Show that $\mu_s = \tan(\text{angle})$



Inclined Plane

□ Newton 2nd law:

$$\sum F_x = mg \sin \theta - \mu_s N = 0$$

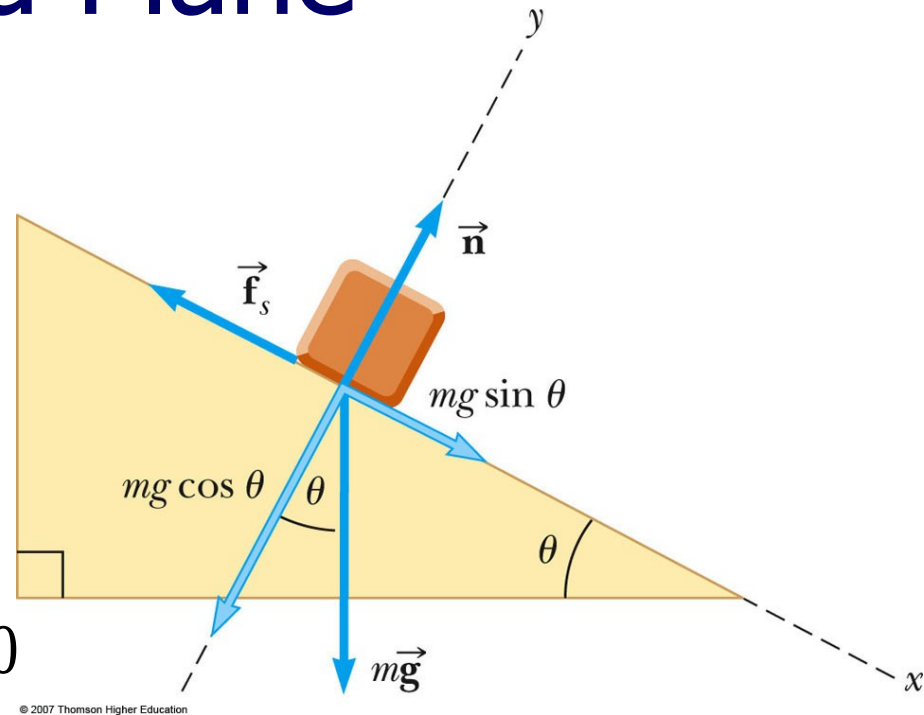
$$\sum F_y = N - mg \cos \theta = 0$$

□ Then $N = mg \cos \theta$

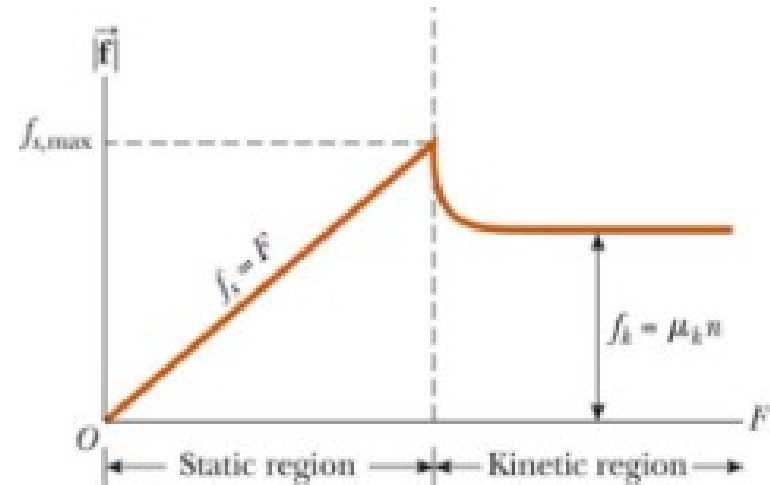
$$\sum F_x = mg \sin \theta - \mu_s mg \cos \theta = 0$$

□ So $\tan \theta = \mu_s = 0.350$

$$\theta = \tan^{-1}(0.350) = 19.3^\circ$$

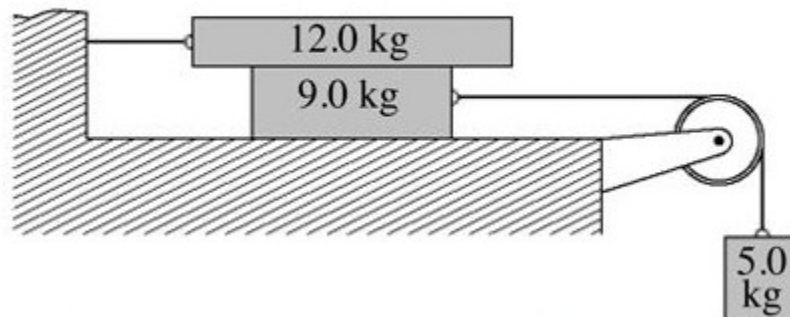


Tricky problem with friction



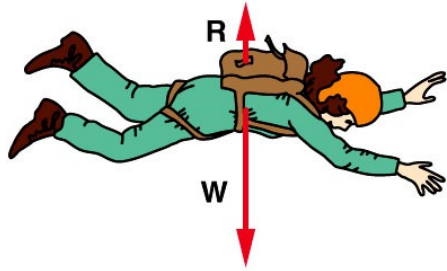
- 2) A 50-kg box is resting on a horizontal floor. A force of 250 N directed at an angle of 30.0° below the horizontal is applied to the box. The coefficient of static friction between the box and the surface is 0.40, and the coefficient of kinetic friction is 0.30. What is the force of friction on the box?

accelerates downward when it is released, find its acceleration.



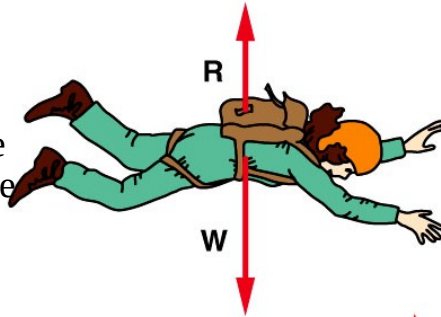
Whats happening?





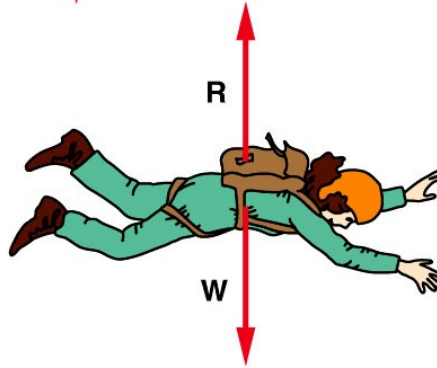
$W > R$
 $F_{\text{net}} @ \text{down}$
 $0 < a < 10 \text{ m/s}^2$
 Speeding up a @down
 $V @ \text{down}$

$W > R$ R gets bigger
 $F_{\text{net}} @ \text{down}$ smaller than before
 $0 < a < 10 \text{ m/s}^2$ smaller than before
 Speeding up a @down
 $V @ \text{down}$

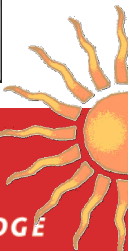
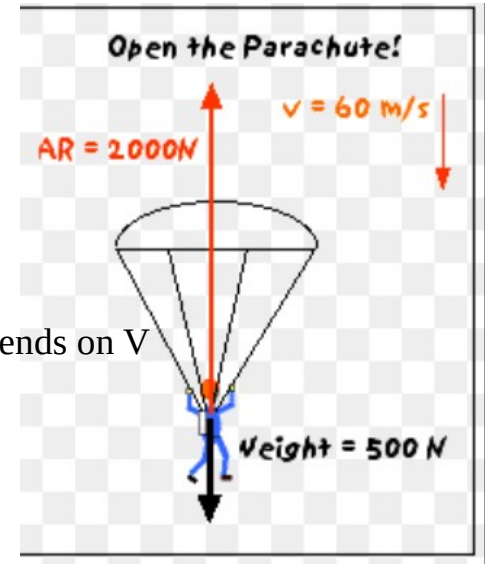


Air resistance depends on
 the speed²
 Faster → more drag

$R = W$
 $F_{\text{net}} = 0$
 $a = 0$ constant velocity
 Terminal speed is reached
 $V @ \text{down}$ constant



$R > W$ but will decrease since depends on V
 $F_{\text{net}} @ \text{up}$
 $0 < a < 10 \text{ m/s}^2$ but @ up
 Slowing down
 $V @ \text{down}$





She opens her parachute. $\text{Drag} > \text{weight}$
 Now the net force is @ up so the acceleration is @ up
 While she is moving @ down.
 So she is slowing down and is still moving @ down.
 The acceleration is negative.
 Drag is increasing



$\text{Drag} = \text{weight}$
 Net force is 0
 Acceleration is 0
 She is going down at a constant speed.
 (terminal speed)
 She is about to touch ground.



TERMINAL SPEEDS :

(from 2007 edition book from same author,
Richard A. Muller)

Food dropped from airplane (3000m)

Example: Afghanistan 2002.

9mph

A person: 74mph

(see 1000 ways to die, you can survive)

With parachute = 13mph

otherwise about 540 mph without air resistance

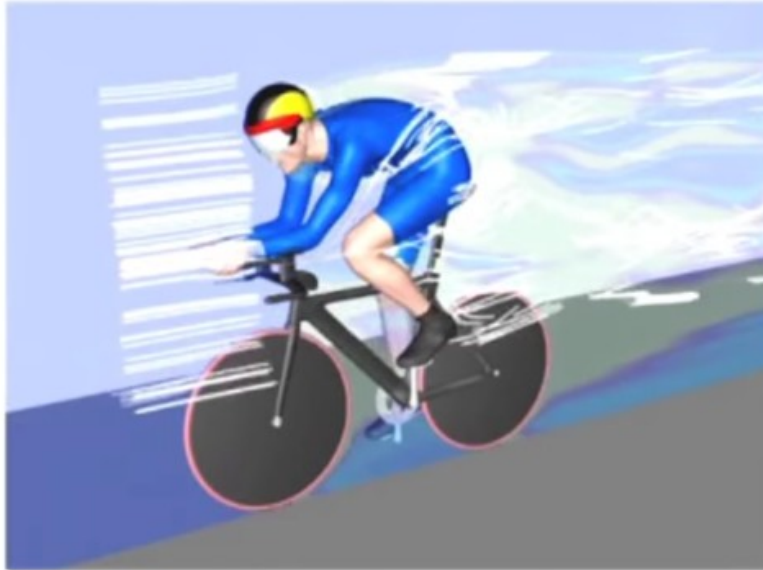
You can use the equation $v^2 = 2 d g$ to compute

This free-fall speed (from m/s to mph multiply by



Physics of sport:
Drag is another type of friction

Drag balances the thrust → no acceleration



$$F = \frac{1}{2} \rho v^2 C_d A$$

Aim = Small F

Drafting

small v, low ρ

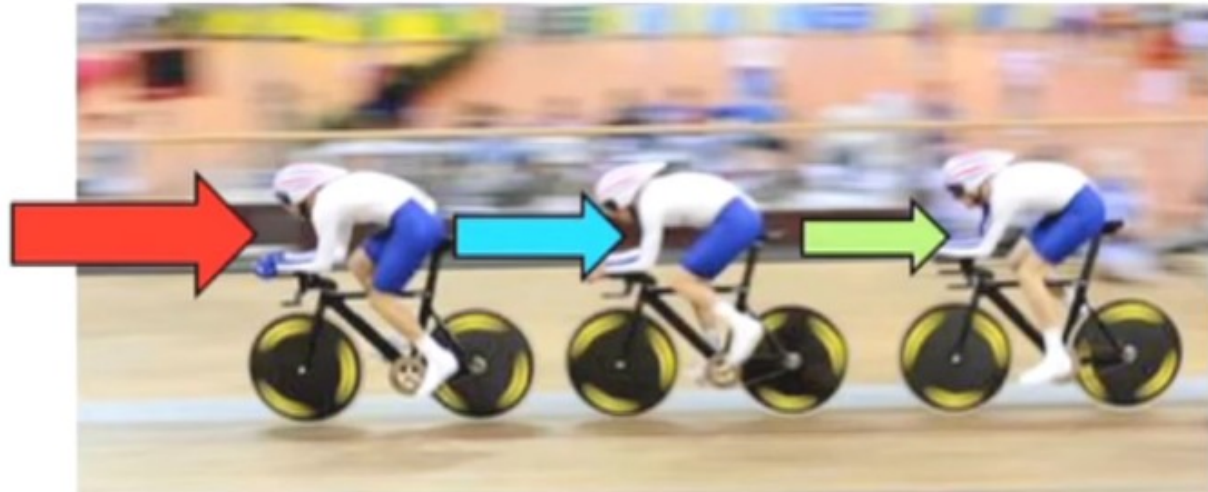
Design

optimisation

Small area, A

Low drag

coefficient, C_d



Hints for Problem-Solving

- ❑ **Read** the problem carefully at least once
- ❑ **Draw** a picture of the system, identify the object of primary interest, and indicate forces with arrows
- ❑ **Label** each force in the picture in a way that will bring to mind what physical quantity the label stands for (e.g., T for tension)
- ❑ **Draw** a free-body diagram of the object of interest, based on the labeled picture. If additional objects are involved, draw separate free-body diagram for them
- ❑ **Choose a convenient coordinate system** for each object
- ❑ **Apply Newton's second law.** The x- and y-components of Newton second law should be taken from the vector equation and written individually. This often results in two equations and two unknowns
- ❑ **Solve** for the desired unknown quantity, and substitute the numbers

$$F_{net,x} = ma_x$$

$$F_{net,y} = ma_y$$



Objects in Equilibrium

- Objects that are either at rest or moving with constant velocity are said to be in **equilibrium**
- Acceleration of an object can be modeled as zero: $\vec{a}=0$
- Mathematically, the net force acting on the object is zero $\sum \vec{F}=0$
- Equivalent to the set of component equations given by

$$\sum F_x = 0$$

$$\sum F_y = 0$$



Accelerating Objects

- If an object that can be modeled as a particle experiences an acceleration, there must be a nonzero net force acting on it
- Draw a free-body diagram
- Apply Newton's Second Law in component form

$$\sum \vec{F} = m\vec{a}$$

$$\sum F_x = ma_x$$

$$\sum F_y = ma_y$$



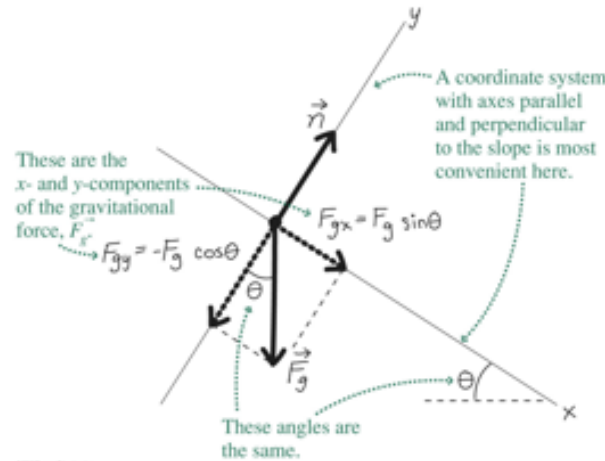
Inclined planes

A Typical Problem: What's the skier's acceleration? What's the force the snow exerts on the skier?

- Physical diagram:



- Free-body diagram:



- Newton's law: $\vec{F}_{\text{net}} = \vec{n} + \vec{F}_g = m\vec{a}$
- In components:
 - x-component: $mg \sin \theta = ma$
 - y-component: $n - mg \cos \theta = 0$
- Solve (with $m = 65 \text{ kg}$ and $\theta = 32^\circ$) to get the answers:

What is the acceleration
What is the normal force?

Textbook 5.4 constant speed
Textbook 5.10 acceleration
516 and 5.17

If there is friction $\rightarrow$ find a with $\mu_k = 0.03$
 $mg \sin(\text{angle}) - F_k = ma$

October 10, 2025

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THE EDGE IN KNOWLEDGE



NEWTON 2 & 3: AN EXAMPLE

A 1-kg and a 2-kg block sit touching on a frictionless surface. You push on the 1-kg block with a 6-N force. What force does the 2-kg block exert on the 1 kg block?

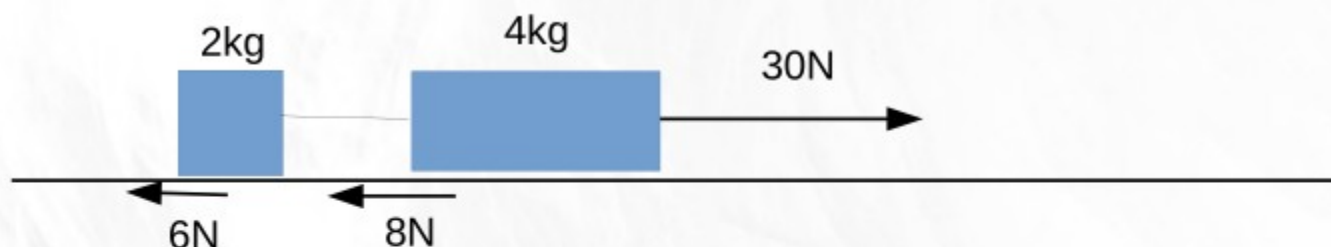


Total mass is 3 kg; with 6 N applied force, Newton 2 ($F = ma$) gives $a = 2 \text{ m/s}^2$ for the acceleration of the pair.



Blocks

Textbook 5.11



2 blocks tied together by a horizontal string are being pulled across the table by a horizontal force of 30N as shown. The 2 kg block has a 6N frictional force exerted on it by the table, and the 4kg block has an 8N frictional force acting on it.

A) What is the net force acting on the entire two-block system ?

(take the whole 6kg system and compute right – left = F_{net})

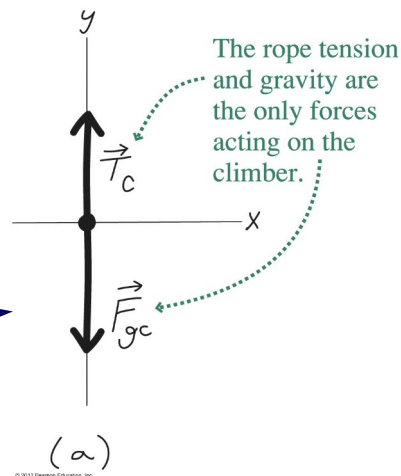
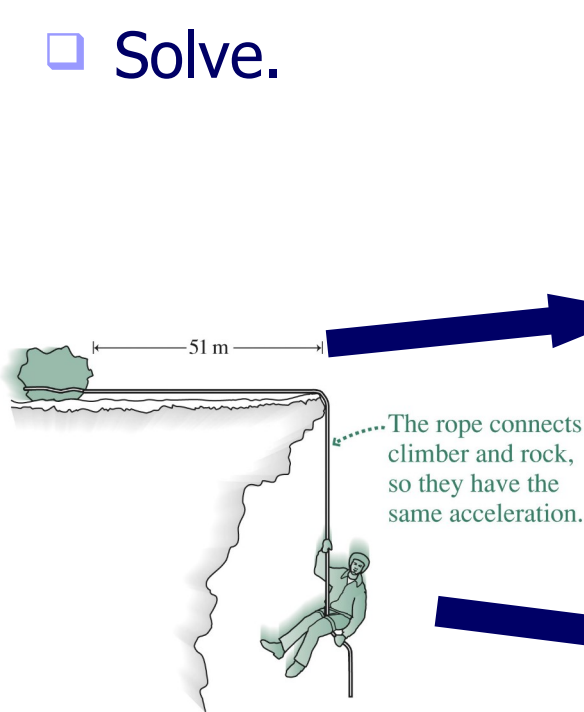
B) What is the acceleration of the system ? ($F_{\text{net}} = ma$ or right – left = ma)

C) What force is exerted on the 2kg block by the connecting string ? (Cut out this 2kg block and consider only the forces acting on this block. Its acceleration is the same as that of the entire system).

D) Find the net force acting on the 4 kg and calculate its acceleration. How does this value compare to that found in B) ?



- Solve problems involving multiple objects by first identifying each object and all the forces on it.
- Draw a free-body diagram for each.
- Write Newton's law for each.
- Identify connections between the objects, which give common terms in the Newton's law equations.
- Solve.



$$\text{climber: } \vec{T}_c + \vec{F}_{gc} = m_c \vec{a}_c$$

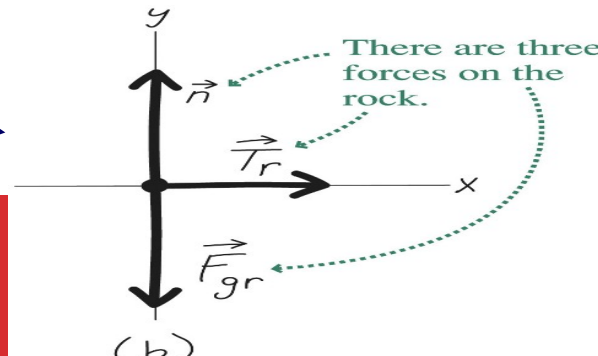
$$\text{rock: } \vec{T}_r + \vec{F}_{gr} + \vec{n} = m_r \vec{a}_r$$

$$\text{climber, y: } T - m_c g = -m_c a$$

$$\text{rock, x: } T = m_r a$$

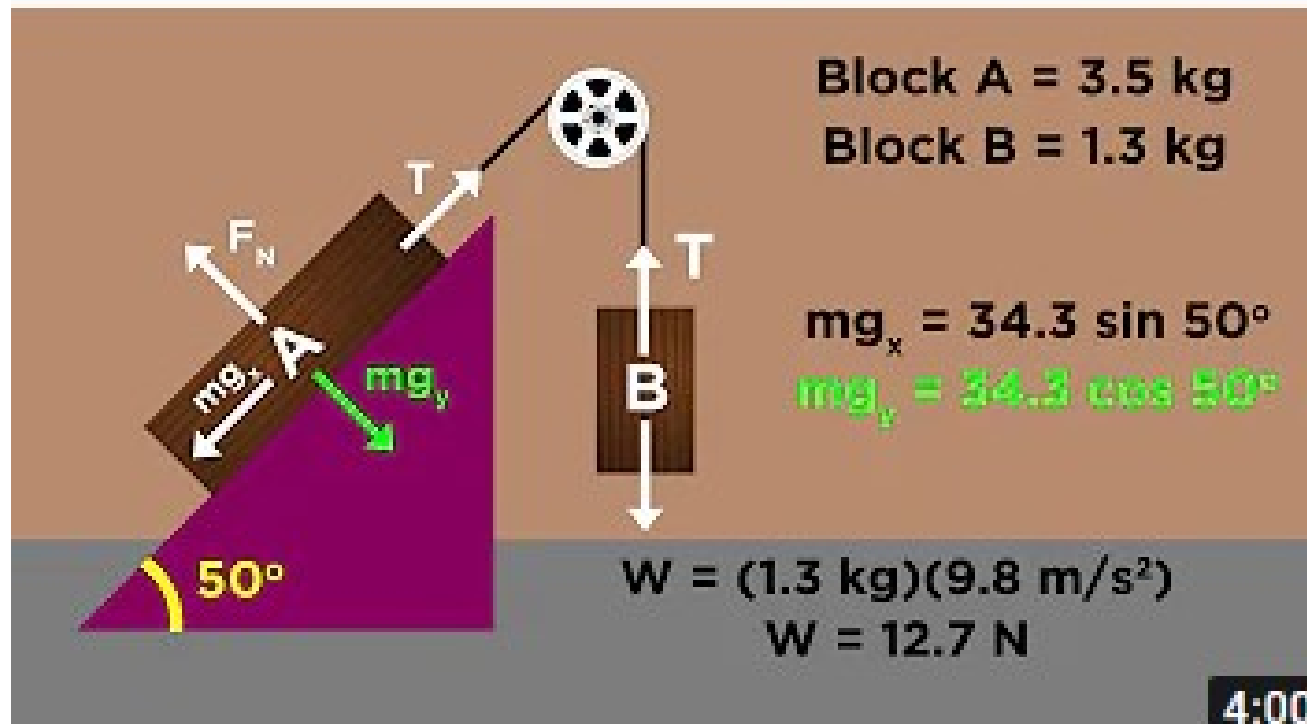
$$\text{rock, y: } n - m_r g = 0$$

$$a = \frac{m_c g}{m_c + m_r}$$



What if you have friction?
μ_k = 0.2





<https://www.youtube.com/watch?v=GOZfbU6tLVk>

Always take the positive direction = direction of acceleration.

If you find $a < 0$. Start over. That means the direction of motion is wrong.

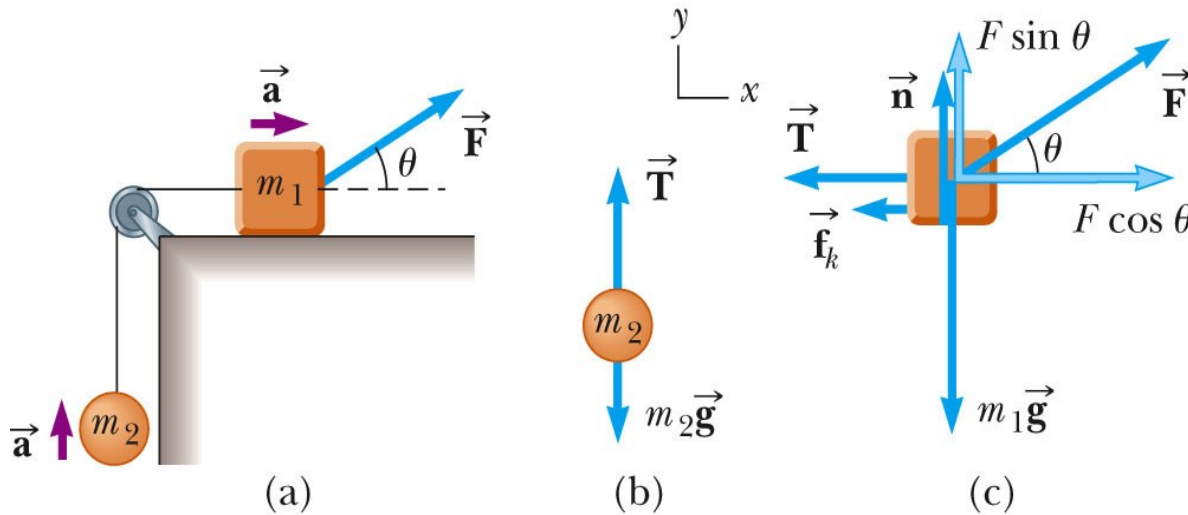
Tutorial youtube in dropbox

<https://www.youtube.com/watch?v=N6IhkTjWrd4>



Textbook 5.5 equ
Textbook 5.12 a

Multiple Objects



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- A block of mass m_1 on a rough, horizontal surface is connected to a ball of mass m_2 by a lightweight cord over a lightweight, frictionless pulley as shown in figure. A force of magnitude F at an angle θ with the horizontal is applied to the block as shown and the block slides to the right. The coefficient of kinetic friction between the block and surface is μ_k . Find the magnitude of acceleration of the two objects.



Multiple Objects

□ m_1 : $\sum F_x = F \cos \theta - f_k - T = m_1 a_x = m_1 a$

$\sum F_y = N + F \sin \theta - m_1 g = 0$

□ m_2 : $\sum F_y = T - m_2 g = m_2 a_y = m_2 a$

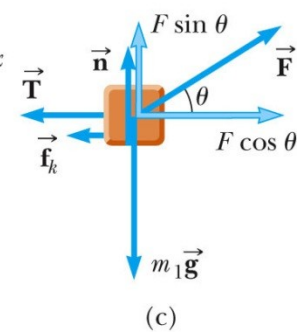
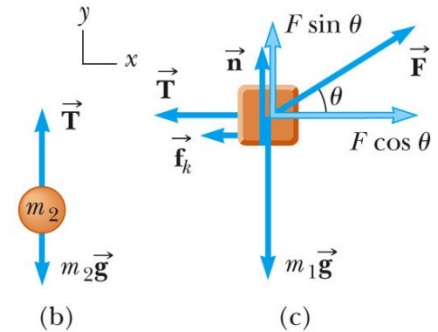
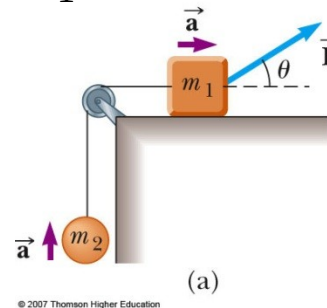
$T = m_2 (a + g)$

$N = m_1 g - F \sin \theta$

$f_k = \mu_k N = \mu_k (m_1 g - F \sin \theta)$

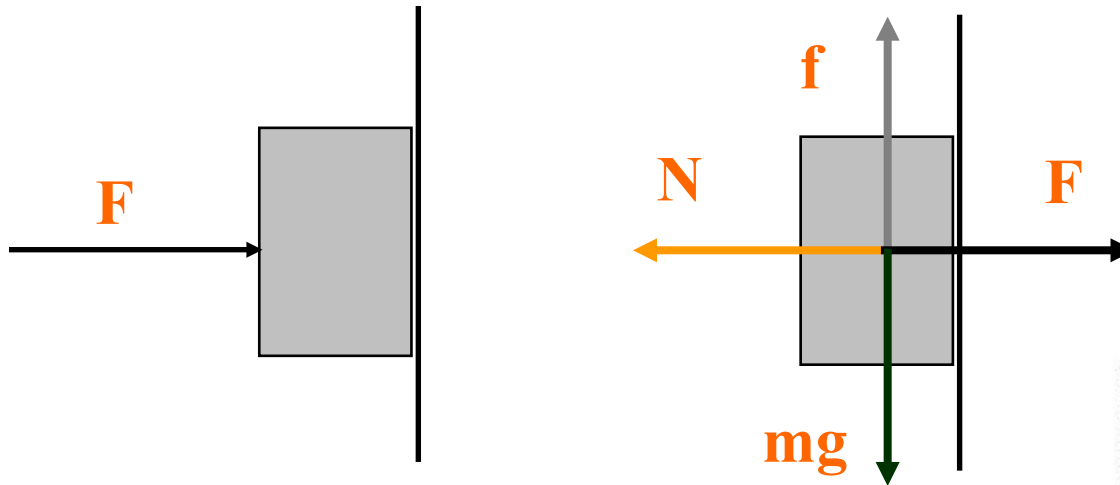
$F \cos \theta - \mu_k (m_1 g - F \sin \theta) - m_2 (a + g) = m_1 a$

$$a = \frac{F (\cos \theta + \mu_k \sin \theta) - (m_2 + \mu_k m_1) g}{m_1 + m_2}$$



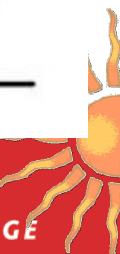
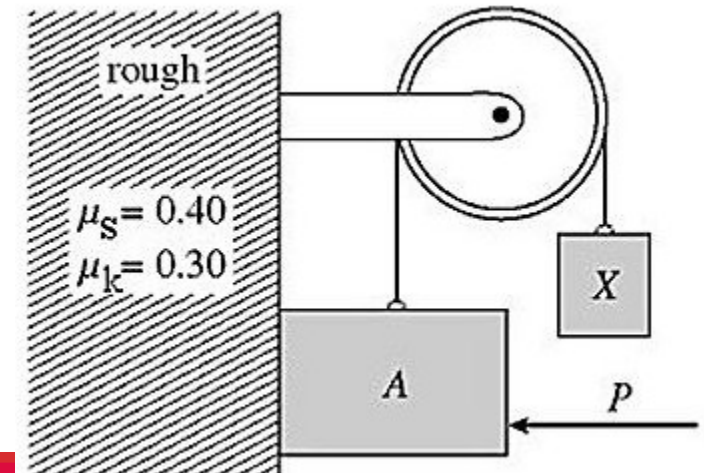
Equilibrium, Example 1

- What is the smallest value of the force F such that the 2.0-kg block will not slide down the wall? The coefficient of static friction between the block and the wall is 0.2. ?

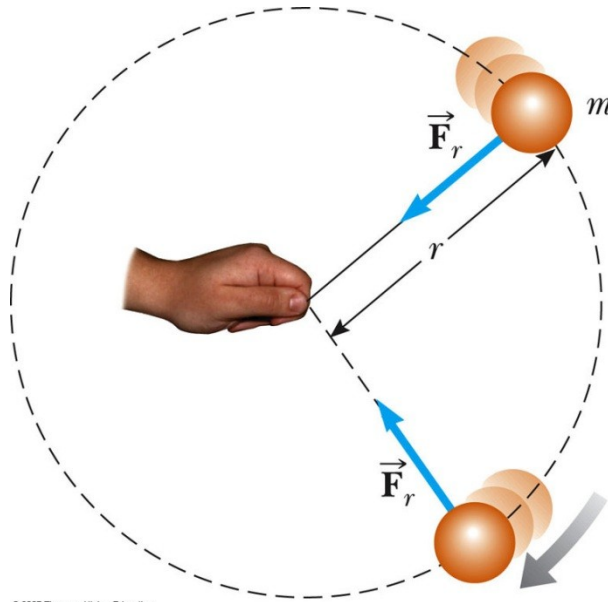


When F is small, there is kinetic friction
And the block slides. $F = N$
When F reaches a threshold,
Static friction = μ_s Normal

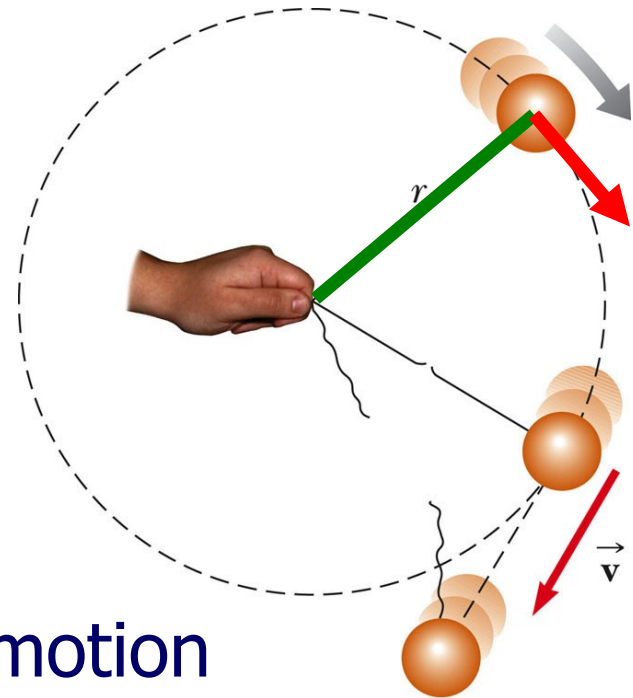
If A slides up and X slides down.
Suppose there is an acceleration.
Draw the forces.



Uniform Circular Motion: Definition



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Uniform circular motion

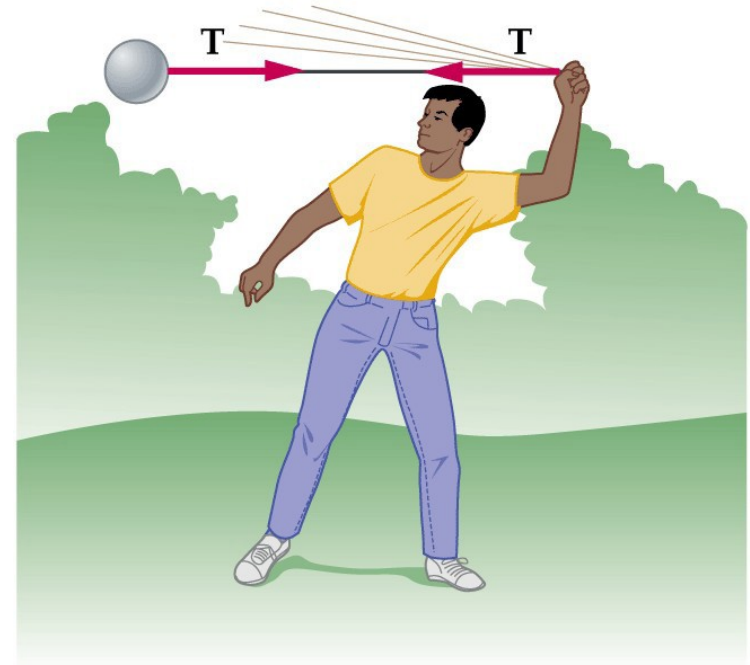
Constant speed, or,
constant magnitude of velocity

Motion along a circle:
Changing direction of velocity



Uniform Circular Motion: Observations

- ❑ Object moving along a curved path with **constant speed**
 - Magnitude of velocity: same
 - Direction of velocity: changing
 - Velocity $\vec{v}$: changing
 - Acceleration is NOT zero!
 - **Net force acting on an object is NOT zero**
 - “Centripetal force”



$$\vec{F}_{net} = m \vec{a}$$



Uniform Circular Motion

□ Magnitude:

$$\Delta \vec{r} = \vec{r}_f - \vec{r}_i$$

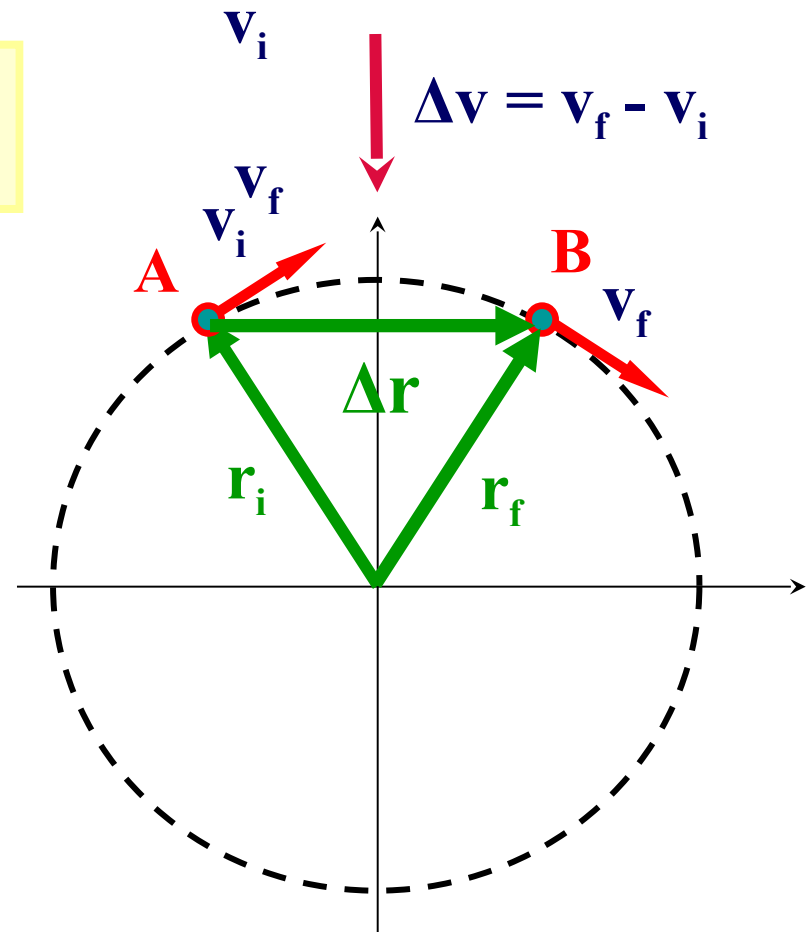
$$\vec{a} = \frac{\Delta \vec{v}}{\Delta t} = \frac{\vec{v}_f - \vec{v}_i}{\Delta t}$$

$$\frac{\Delta v}{v} = \frac{\Delta r}{r} \quad \text{so,} \quad \Delta v = \frac{v \Delta r}{r}$$

$$\frac{\Delta v}{\Delta t} = \frac{\Delta r}{\Delta t} \frac{v}{r} = \frac{v^2}{r}$$

$$a_r = \frac{\Delta v}{\Delta t} = \frac{v^2}{r}$$

□ Direction: **Centripetal**



Uniform Circular Motion

□ Velocity:

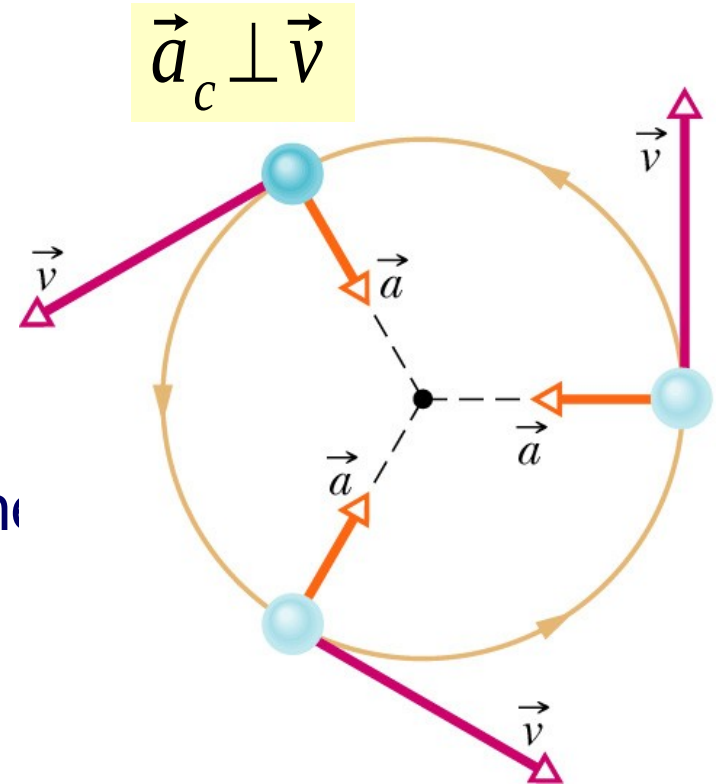
- Magnitude: constant v
- The direction of the velocity is tangent to the circle

□ Acceleration:

- Magnitude: $a_c = \frac{v^2}{r}$
- directed toward the center of the circle of motion

□ Period:

- time interval required for one complete revolution of the particle



$$T = \frac{2\pi r}{v}$$



Centripetal Force

□ Acceleration:

- Magnitude:

$$a_c = \frac{v^2}{r}$$

- Direction: toward the center of the circle of motion

□ Force:

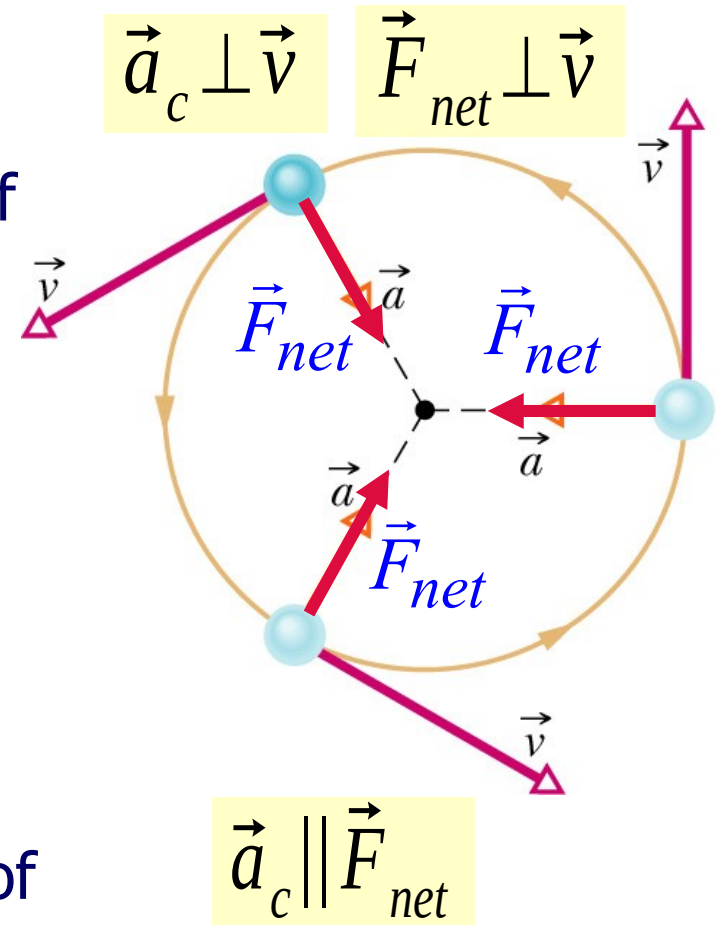
- Start from Newton's 2nd Law

$$\vec{F}_{net} = m\vec{a}$$

- Magnitude:

$$F_{net} = ma_c = \frac{mv^2}{r}$$

- Direction: toward the center of the circle of motion



What provides Centripetal Force ?

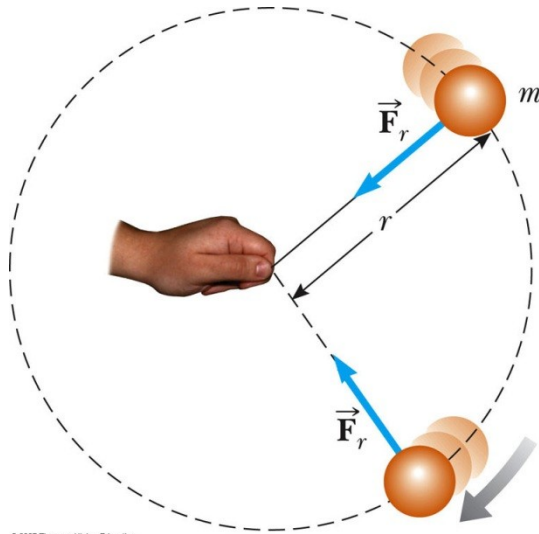
- ❑ Centripetal force is not a new kind of force
- ❑ Centripetal force refers to any force that keeps an object following a circular path

$$F_c = ma_c = \frac{mv^2}{r}$$

- ❑ Centripetal force is a combination of
 - Gravitational force mg : downward to the ground
 - Normal force $\mathbf{N}$: perpendicular to the surface
 - Tension force $\mathbf{T}$: along the cord and away from object
 - Static friction force: $f_s^{max} = \mu_s N$



What provides Centripetal Force ?

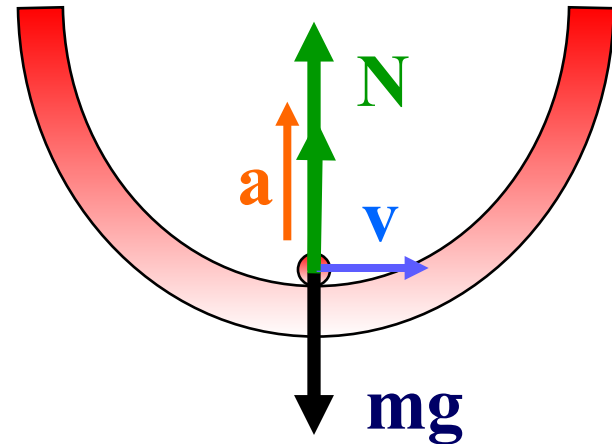


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$$F_{net} = T = ma$$
$$T = \frac{mv^2}{r}$$

$$F_{net} = N - mg = ma$$

$$N = mg + m \frac{v^2}{r}$$

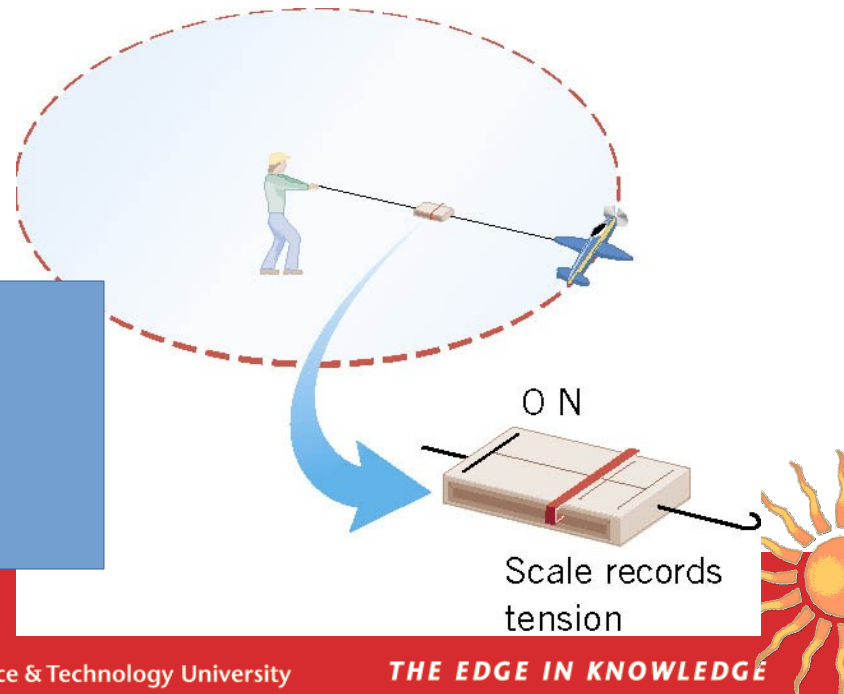


5.3 Centripetal Force

Example 5: The Effect of Speed on Centripetal Force

The model airplane has a mass of 0.90 kg and moves at constant speed on a circle that is parallel to the ground. The path of the airplane and the guideline lie in the same horizontal plane because the weight of the plane is balanced by the lift generated by its wings. Find the tension in the 17 m guideline for a speed of 19 m/s.

$$F_c = T = m \frac{v^2}{r}$$



Conceptual Example 6: A Trapeze Act

In a circus, a man hangs upside down from a trapeze, legs bent over and arms downward, holding his partner. Is it harder for the man to hold his partner when the partner hangs straight down and is stationary or when the partner is swinging through the straight-down position?



**Discuss the
Free body diagram**



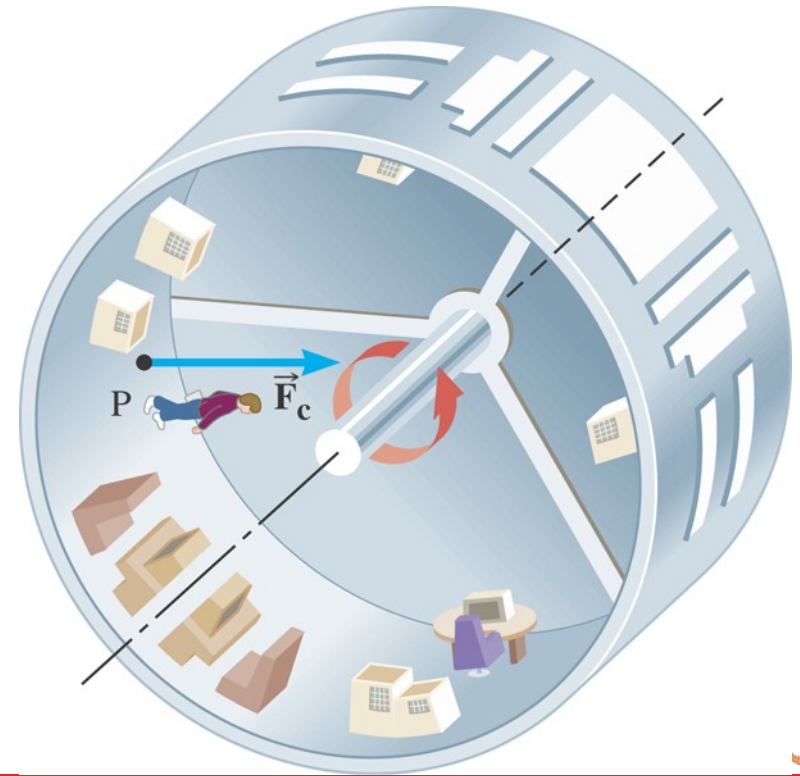
5.6 Apparent Weightlessness and Artificial Gravity

Example 13: Artificial Gravity → discuss centripetal vs centrifugal

At what speed must the surface of the space station move so that the astronaut experiences a push on his feet equal to his weight on earth? The radius is 1700 m.

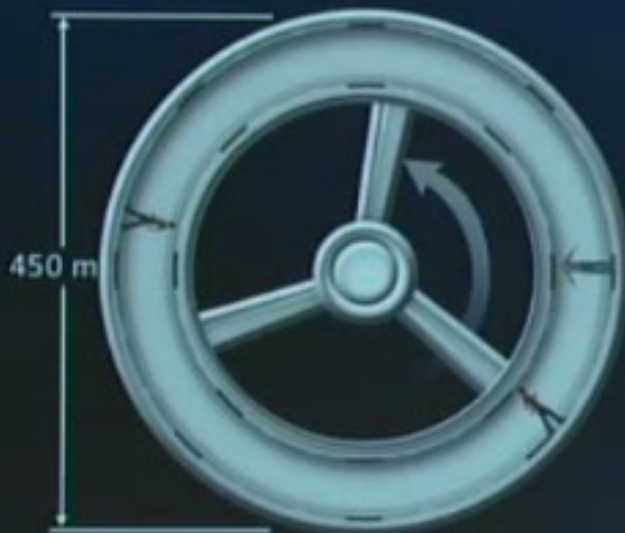
$$F_c = m \frac{v^2}{r} = mg$$

$$v = \sqrt{rg}$$
$$= \sqrt{(1700 \text{ m})(9.80 \text{ m/s}^2)}$$



“Artificial Gravity”

Spin a space station so that an astronaut’s centripetal acceleration is equal to that of gravity at Earth’s surface. If a space station is in the shape of a ring 450 meters in diameter, at what rate must it spin?



Want: $\frac{v^2}{r} = g$, the acceleration of gravity

Solve: $v = \sqrt{rg}$

Time T for 1 revolution: $T = 2\pi r / v$

Substitute for v : $T = 2\pi r / \sqrt{rg} = 2\pi \sqrt{r/g}$

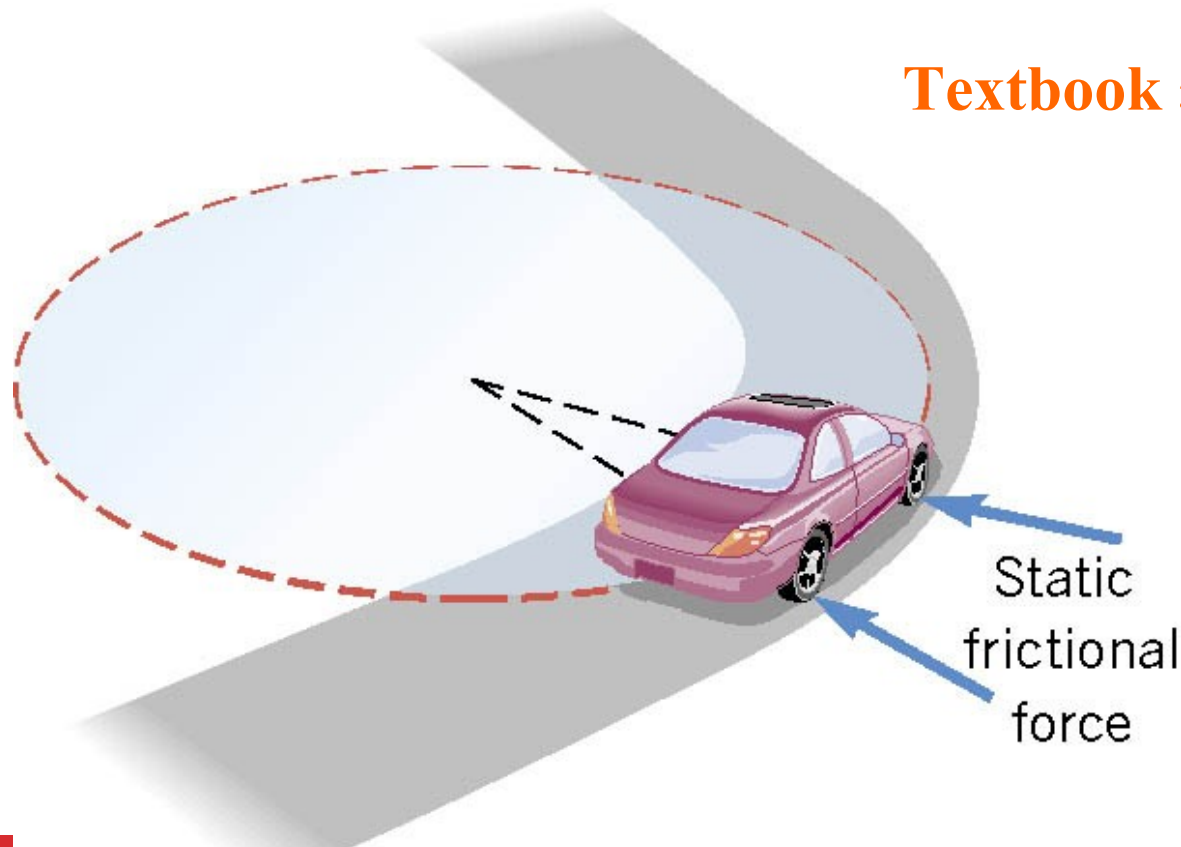
Do the numbers: $r = 225 \text{ m}$, $g = 9.8 \text{ m/s}^2$



5.4 Banked Curves

On an unbanked curve, the static frictional force provides the centripetal force.

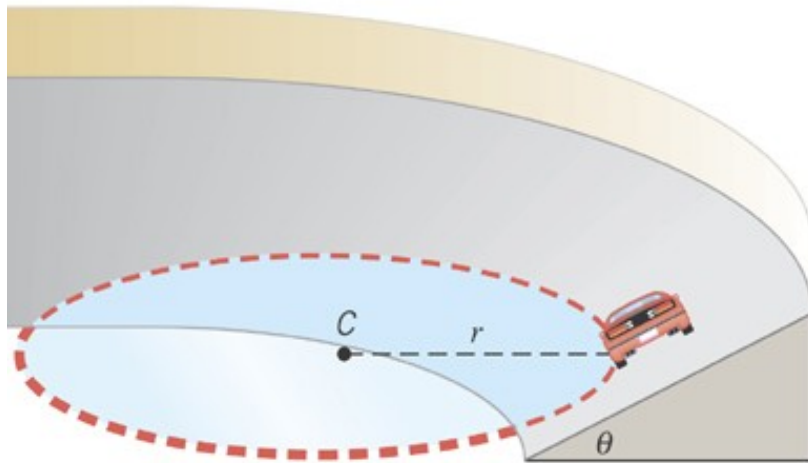
Textbook 5.21



5.4 Banked Curves

On a frictionless banked curve, the centripetal force is the horizontal component of the normal force. The vertical component of the normal force balances the car's weight.

→ you don't take the + x-axis along the slope but toward the Center In the same direction as the acceleration



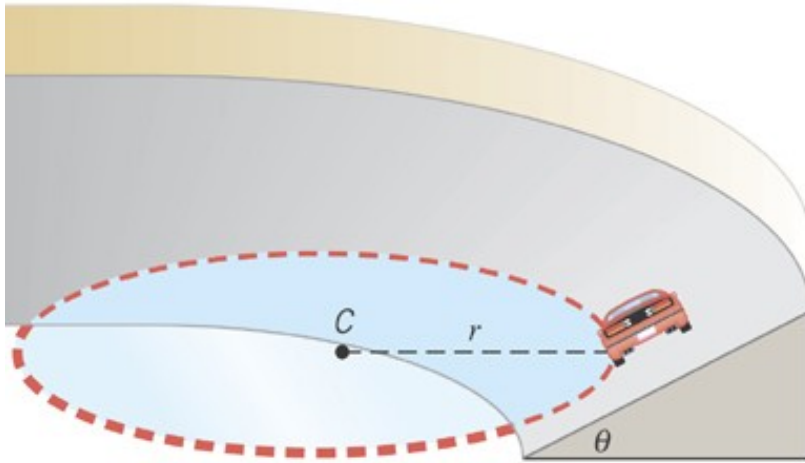
(a)

**Draw the forces
Acting on the car.
Acceleration is in
+ is in**

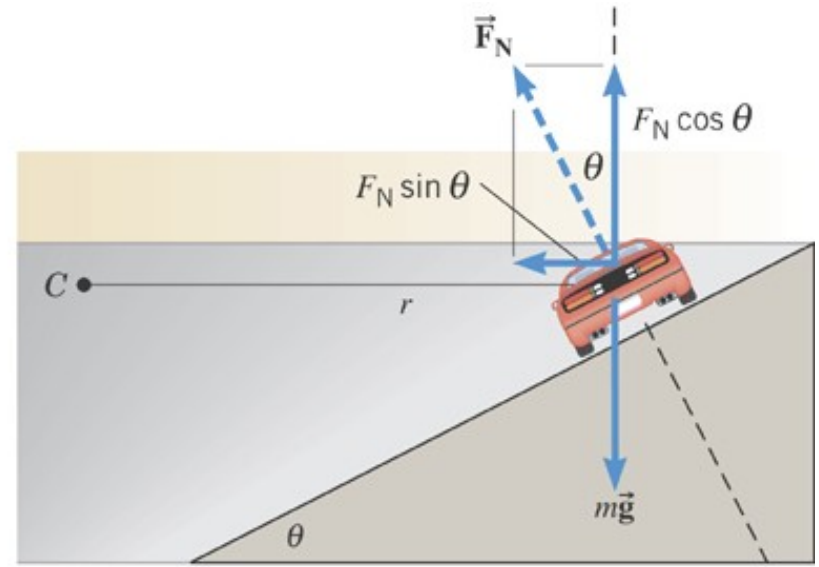
(b)



5.4 Banked Curves



(a)



(b)

$$F_c = F_N \sin \theta = m \frac{v^2}{r}$$

$$F_N \cos \theta = mg$$



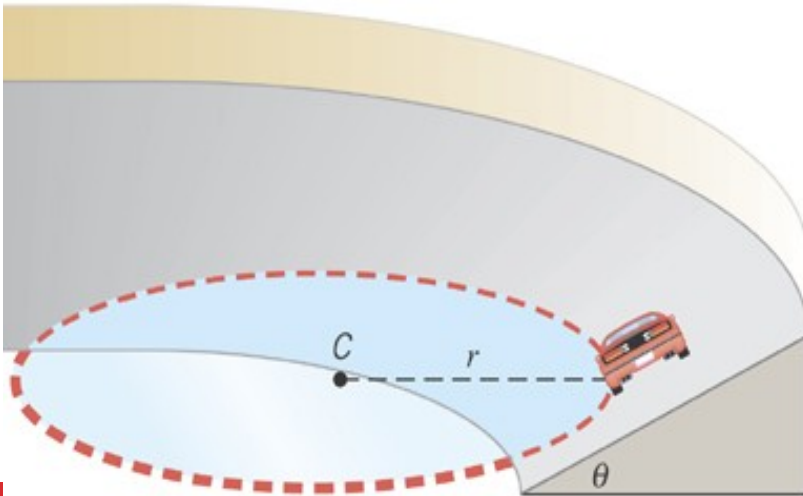
5.4 Banked Curves

$$F_N \sin \theta = m \frac{v^2}{r}$$

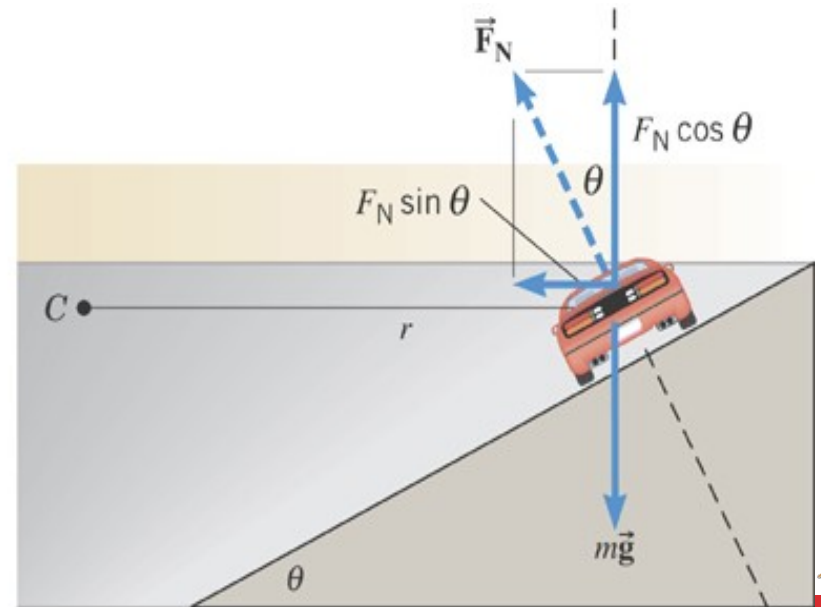


$$\tan \theta = \frac{v^2}{rg}$$

$$F_N \cos \theta = mg$$



(a)



(b)



5.4 Banked Curves

Example 8: The Daytona 500

The turns at the Daytona International Speedway have a maximum radius of 316 m and are steely banked at 31 degrees. Suppose these turns were frictionless. As what speed would the cars have to travel around them?



5.4 Banked Curves

Example 8: The Daytona 500

The turns at the Daytona International Speedway have a maximum radius of 316 m and are steely banked at 31 degrees. Suppose these turns were frictionless. As what speed would the cars have to travel around them?

$$\tan \theta = \frac{v^2}{rg} \quad \longrightarrow \quad v = \sqrt{rg \tan \theta}$$

$$v = \sqrt{(316 \text{ m})(9.8 \text{ m/s}^2) \tan 31^\circ} = 43 \text{ m/s} \quad (96 \text{ mph})$$

Textbook 5.21



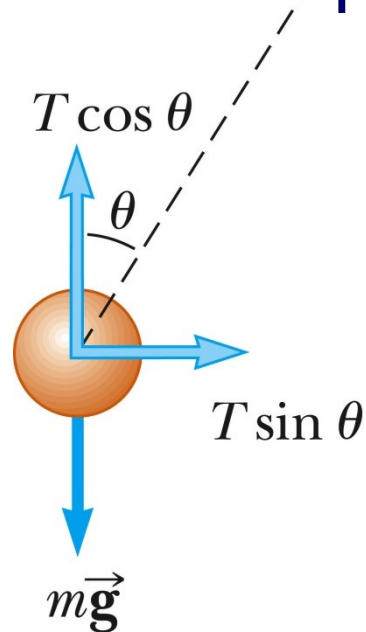
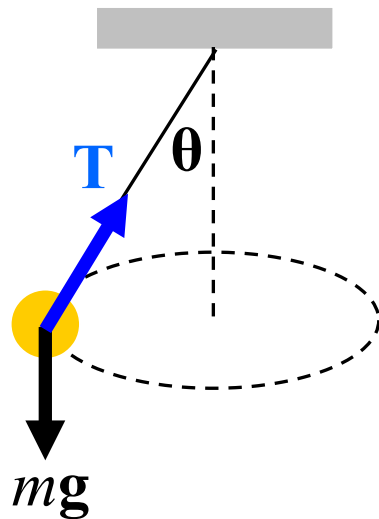
Problem Solving Strategy

- ❑ **Draw a free body diagram**, showing and labeling all the forces acting on the object(s)
- ❑ **Choose a coordinate system** that has one axis perpendicular to the circular path and the other axis tangent to the circular path
- ❑ **Find the net force toward the center** of the circular path (this is the force that causes the centripetal acceleration, F_c)
- ❑ **Use Newton's second law**
 - The directions will be radial, normal, and tangential
 - The acceleration in the radial direction will be the centripetal acceleration
- ❑ **Solve for the unknown(s)**



The Conical Pendulum

- A small ball of mass $m = 5$ kg is suspended from a string of length $L = 5$ m. The ball revolves with constant speed v in a horizontal circle of radius $r = 2$ m. Find an expression for v and a .

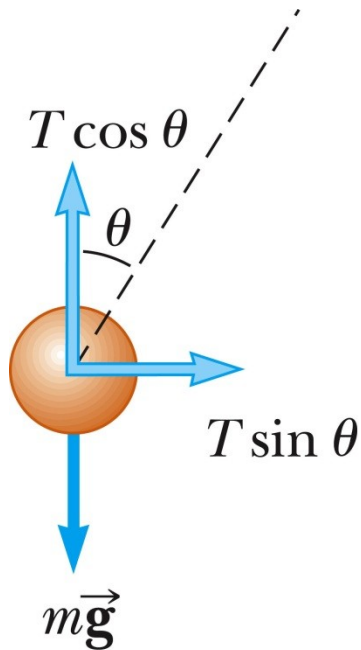


Textbook 5.20



The Conical Pendulum

□ Find v and a



(b)

$$m = 5 \text{ kg} \quad L = 5 \text{ m} \quad r = 2 \text{ m}$$

$$\sum F_y = T \cos \theta - mg = 0$$

$$T \cos \theta = mg$$

$$\sum F_x = T \sin \theta = \frac{mv^2}{r}$$

$$\sin \theta = \frac{r}{L} = 0.4$$

$$\tan \theta = \frac{r}{\sqrt{L^2 - r^2}} = 0.44$$

$$T \sin \theta = \frac{mv^2}{r}$$

$$T \cos \theta = mg$$

$$\tan \theta = \frac{v^2}{gr}$$

$$v = \sqrt{rg \tan \theta}$$

$$v = \sqrt{Lg \sin \theta \tan \theta} = 2.9 \text{ m/s}$$

$$a = \frac{v^2}{r} = g \tan \theta = 4.3 \text{ m/s}^2$$

What is the period of the motion? The tension ?





Textbook 5.35



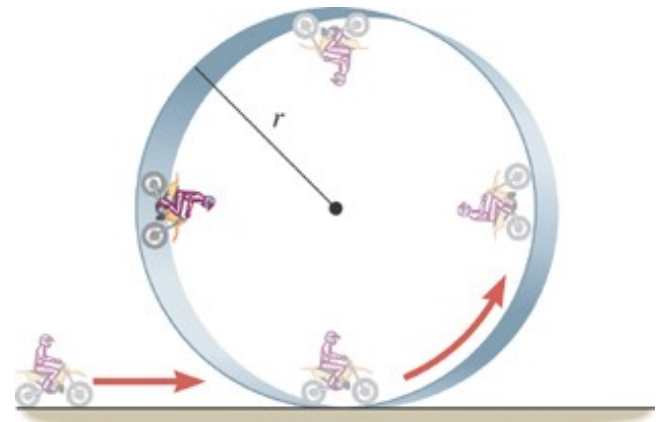
5.7 Vertical Circular Motion

-Draw the forces
Acting on the biker
At each point.

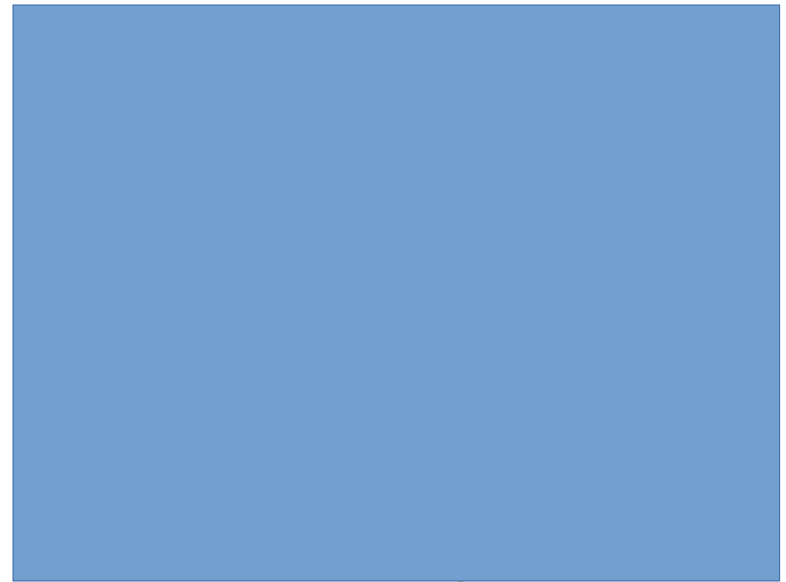
Remember take the
Positive x-axis toward the
Center. Same direction as
acceleration

- At the top. If $N = 0$ the bike
Leaves the track $\rightarrow$ Falling. $\rightarrow$
Minimum speed to make it
Around.

- When $v > v_{\min}$ $N > 0$



(a)



(b)



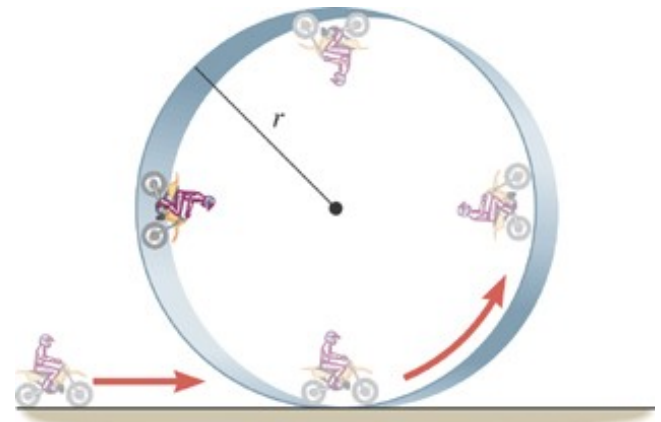
5.7 Vertical Circular Motion

$$F_{N1} - mg = m \frac{v_1^2}{r}$$

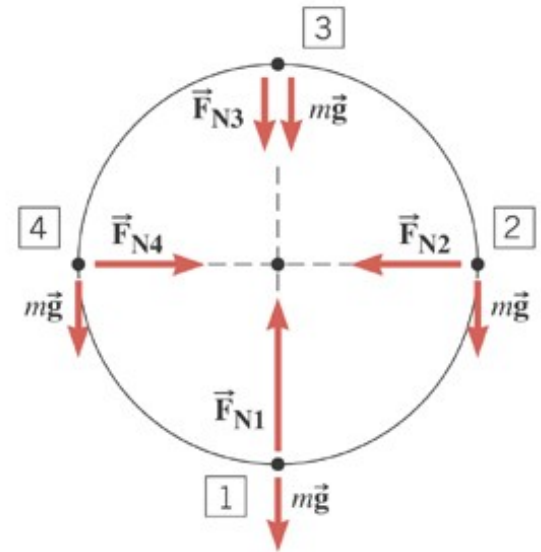
$$F_{N2} = m \frac{v_2^2}{r}$$

$$F_{N4} = m \frac{v_4^2}{r}$$

$$F_{N3} + mg = m \frac{v_3^2}{r}$$



(a)



(b)

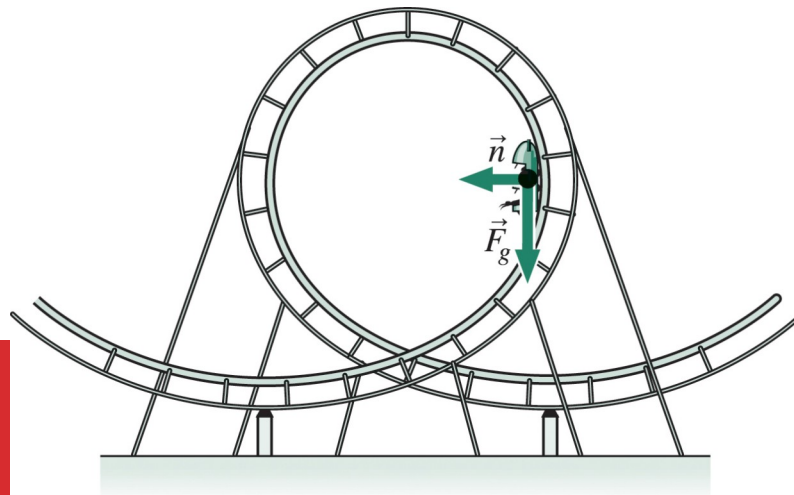


Loop-the-Loop!

- ❑ The two forces acting on the roller-coaster car are:
 - gravity
 - normal force
- ❑ Gravity is always downward, and the normal force is perpendicular to the track.
- ❑ At the position shown, the two forces are at right angles:
 - The normal force acts perpendicular to the car's path, keeping its direction of motion changing.
 - Gravity acts opposite the car's velocity, slowing the car.
 - The net force is *not* toward the center

Newton's law :

$$\vec{n} + \vec{F}_g = m\vec{a}$$

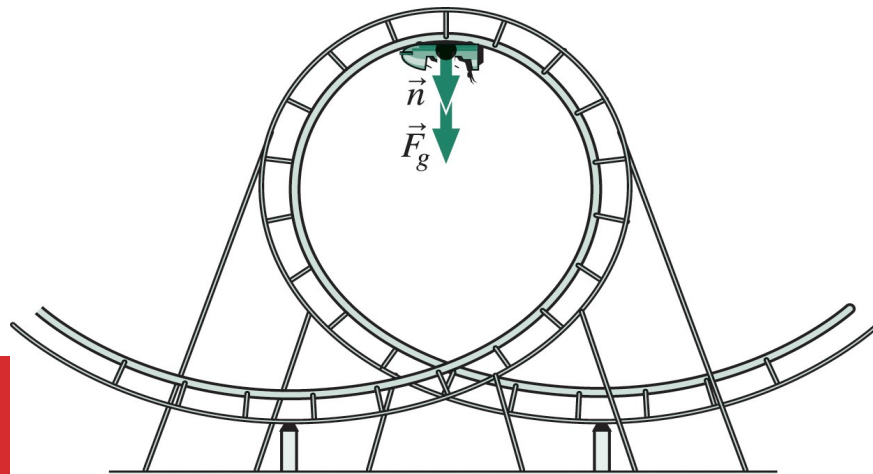


Loop-the-Loop!

- At the top of the loop, both forces are downward:

$$n_y = n, F_{gy} = mg \quad \Rightarrow \quad n + mg = \frac{mv^2}{r}$$

- Solving for v , we obtain $v = \sqrt{nr / m + gr}$
- For the car to stay in contact with the track, the normal force must be greater than zero.
- So the minimum speed is the speed that let the normal force get arbitrarily close to zero at the top of the loop.
- Then gravity alone provides the force that keeps the car in circular motion.

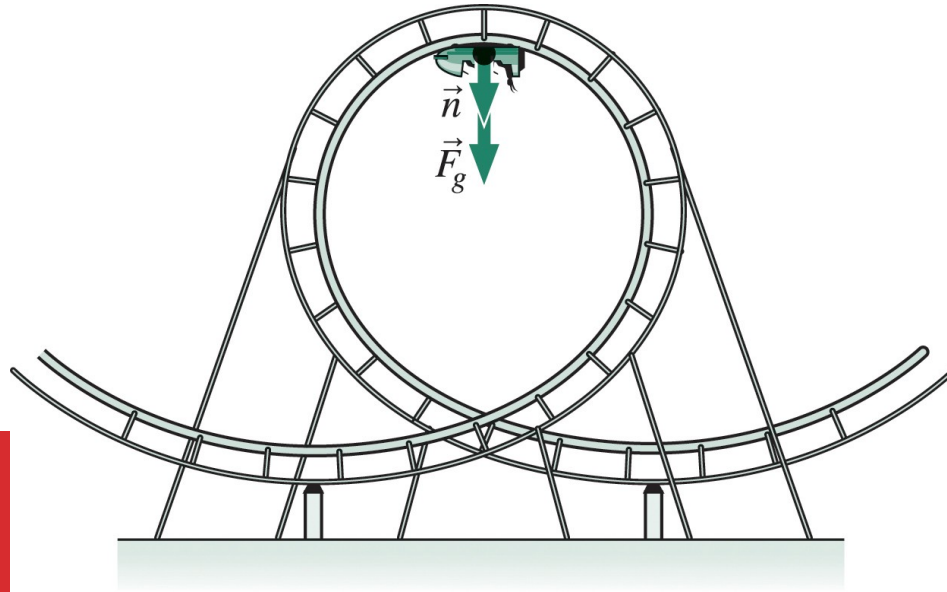


Loop-the-Loop!

- Therefore Newton's law has a single component, with the gravitational force mg providing the acceleration v^2/r that holds the car in its circular path:

$$\vec{F} = m\vec{a} \rightarrow mg = \frac{mv^2}{r}$$

- Solving for the minimum speed at the loop top gives $v = \sqrt{gr}$.
- Slower than this at the top, and the car will leave the track!
- Since this result is independent of mass, car and passengers will all remain on the track as long as $v \geq \sqrt{gr}$.



Loop-the-Loop: How fast?



Doing the numbers:

“The Revolution,”
Six Flags Magic Mountain,
Valencia, California

$r = 6.3$ meters

$$v = \sqrt{gr} = \sqrt{(9.8 \text{ m/s}^2)(6.3 \text{ m})} = 7.9 \text{ m/s} \quad (18 \text{ miles/hour})$$

Textbook 5.23 Ferris wheel



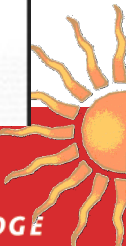
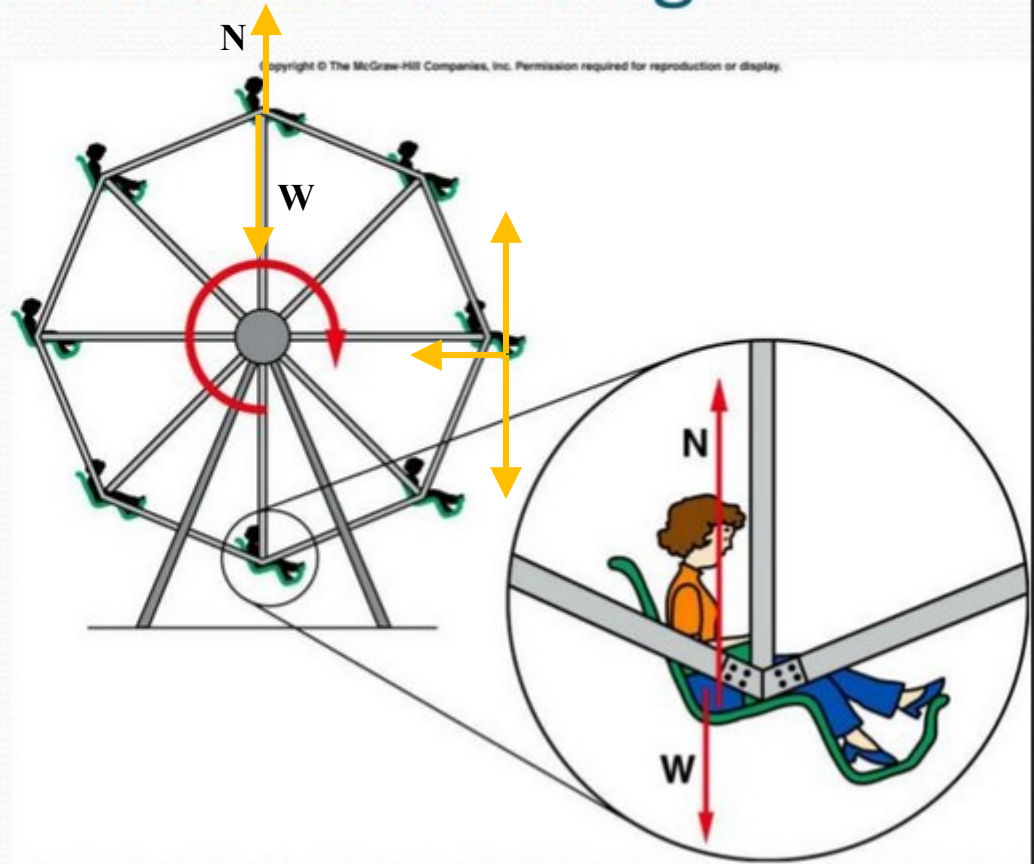
Notice how the force toward the center has to be larger →

In – out = mass x acceleration

What forces are involved in riding a Ferris wheel?

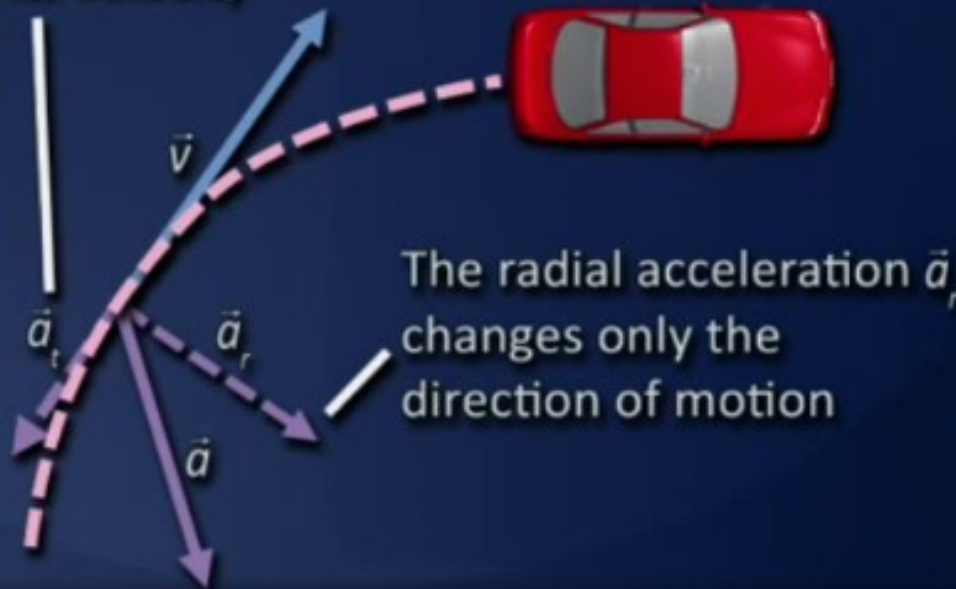
Depending on the position:

- **Weight of the rider**
- **Normal force from seat**
- **Seat-belt force**

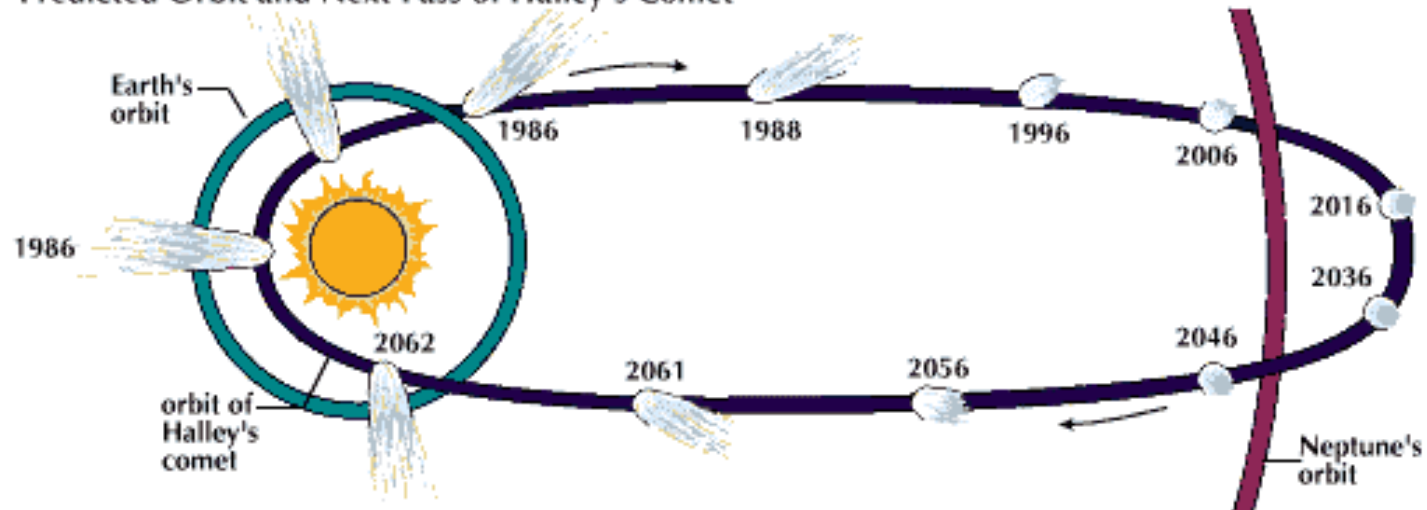


Nonuniform Circular Motion

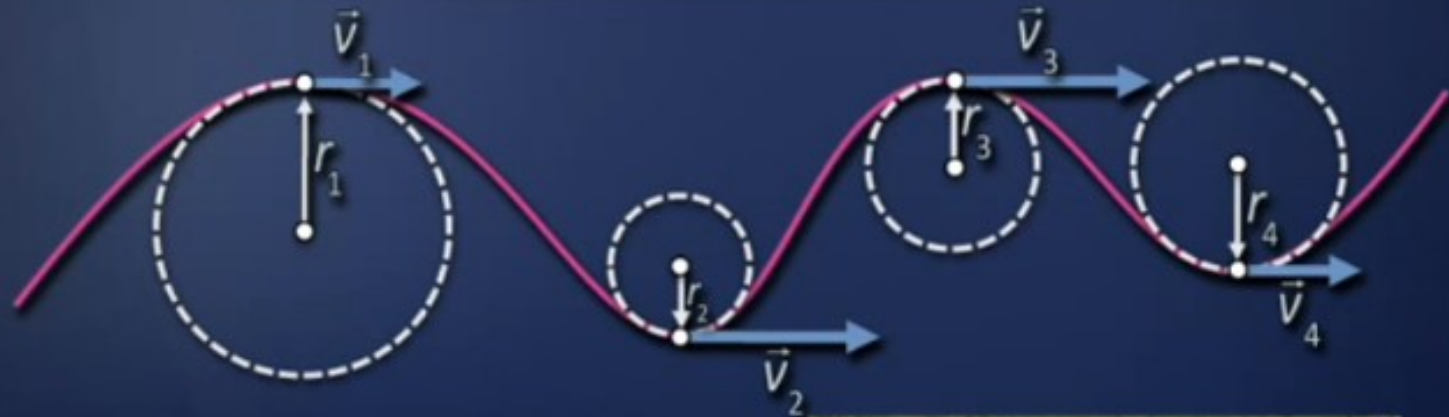
The car is slowing, so its tangential acceleration $\vec{a}_t$ is opposite its velocity



Predicted Orbit and Next Pass of Halley's Comet



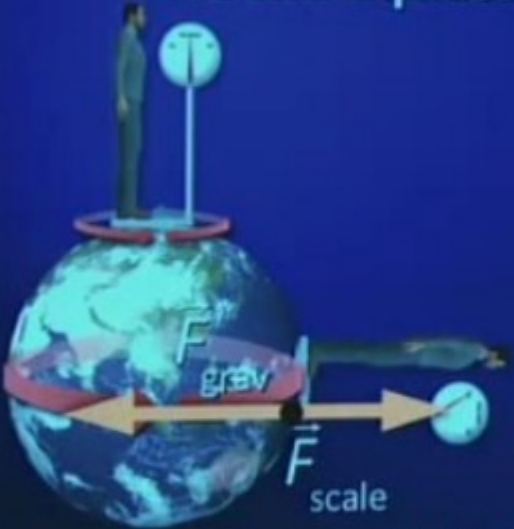
Nonuniform Circular Motion



Osculating circle



At the Equator: What's the scale read?



Acceleration direction?
Toward Earth's center

So net force: toward center

Forces acting?
Gravity (down)
Scale (up)

So gravitational force (actual weight)
is larger than scale force

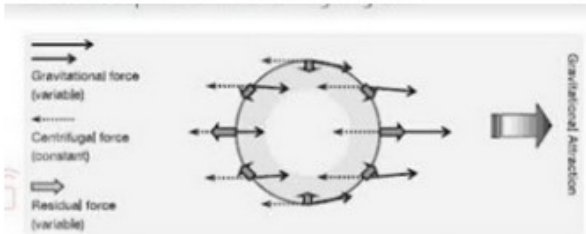
Result: Apparent weight less than actual weight

The normal force is your apparent weight. Always. Reading
From a bathroom scale.



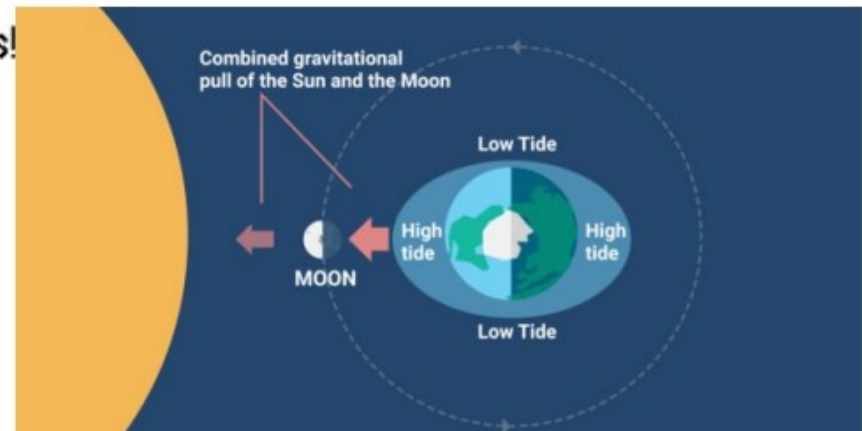
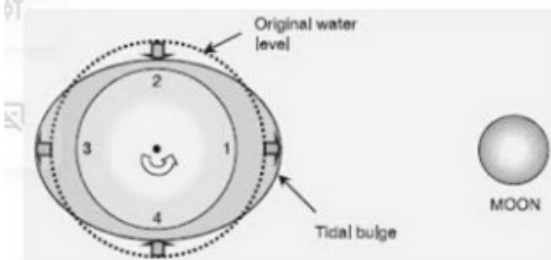
Tides: Differential forces from the Moon (mostly)

- The force by the Moon on the Earth is **greatest** on the **near** side and **smallest** on the **far** side.
- This causes two high tides at each of these points.
- There are **TWO** high tides per 24 hours!



If the Earth were completely covered by a very deep ocean this residual force would act on the water and produce two bulges, one on the side facing the Moon and one on the opposite side of the Earth.

The following diagram shows the Earth and Moon as viewed from above the North Pole. An observer under the water on the Earth at position 1 is beneath one of the water bulges and would be experiencing a high tide. A quarter of a revolution later, at position 2 where the original water level has been depressed, a low tide would be seen. The high tide, low tide sequence repeats as the observer moves around through points 3 and 4.



Spring tides.

These more extreme tides, with a greater tidal range, are called spring tides. Spring tides don't just occur in the spring; they occur whenever the Moon is in a new-moon or full-moon phase, about every 14 days. Spring tides occur when the tidal bulges from the Moon and Sun are aligned.

neap tide The tide of small range that occurs every 14 days, near the times of the first and last quarter of the Moon, when the Moon, Earth, and Sun are at right



Is there a paradox ?

